Stored or Computed? Separating Compositional from Holistic Representations of Binomials in Language Models

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Abstract

Humans balance abstract and item-specific knowledge when ordering binomials: abstract constraints shape the preferences for infrequent binomials, while idiosyncratic knowledge drives them for frequent ones. Evaluating 16 language models across five families, we demonstrate a parallel trade-off. Probing hidden-state representations shows that language models do learn abstract preferences. However, a binomial's relative frequency in the training corpus predicts a model's preference over and above anything those abstract preferences supply, and predicts it more strongly for high-frequency binomials. To locate that item-specific component, we zero out cross-word attention, forcing the representation into a strictly additive function of the two words. The intervention mitigates but does not remove the idiosyncrasy of high-frequency binomials, indicating that the lexical representations themselves come to represent some amount of the phrase-level representation as well. Examining the developmental trajectory across checkpoints further shows that abstract preferences become measurable earlier in training than the frequency-dependent storage does, at a point governed by training tokens rather than model scale.

1 Introduction

Language users are constantly faced with a trade-off between holistically stored, item-specific knowledge and abstract, generative knowledge. For example, the meaning of a phrase often depends on whether it is composed from its individual components or retrieved as a whole. Idioms, such as *kicked the bucket*, are the clearest form of this: the compositional meaning refers to the action of kicking a bucket whereas the idiomatic meaning refers to someone passing away. More generally, the representation of a multi-word expression may be composed from the representations of its individual pieces, or the expression may be stored and retrieved holistically. How, and crucially when, this trade-off emerges in learning has been a question that linguists have been asking for nearly a century (Houghton, 2025a; Houghton and Kapatsinski, 2026; Houghton and Morgan, 2023, 2024; Morgan and Levy, 2015, 2024).

1.1 Abstract Preferences and Holistic Storage

Historically, these questions have been largely addressed at the behavioral level. Berko (1958) famously demonstrated that children could produce the correct plural of nouns that they had never heard before rather early in life; because the plural form was never available to be stored, it must instead have been generated using abstract knowledge of the language. Evidence for abstract knowledge of this kind has been relatively uncontroversial, and has been demonstrated at many different levels of language. The priming literature, for instance, has demonstrated that sentences sharing syntactic structure (e.g., Bock, 1986) or semantic structure (e.g., Meyer and Schvaneveldt, 1971) facilitate the processing of later sentences sharing that structure, which requires that the shared structure be represented independently of the particular words that realize it.

Evidence for holistic storage, on the other hand, has been somewhat more controversial. Nonetheless, evidence of holistic storage has been accumulating over the last few decades (Bybee, 1999; Houghton, 2025b; Kapatsinski and Radicke, 2009; Morgan and Levy, 2015, 2016, 2024; Raymond and Brown, 2012; Raymond et al., 2016; Stemberger and MacWhinney, 2004, 1986). Perhaps the most famous example comes from Bybee and Scheibman (1999). In their seminal study, they demonstrated that the extent to which the word *don't* is reduced is determined by the frequency of the phrase it occurs in. *I don't know* can be reduced to a single nasalized vowel, while *don't* in less fre-

quent phrases, such as *I don't walk*, is unable to be reduced as much. The context-specificity of this reduction demonstrates that the representation of certain high-frequency phrases cannot be reduced to the simple composition of the representations of the individual words.

Binomial expressions (e.g., *cats and dogs*) are notable for their relationship between abstract preferences and holistic storage: they show evidence of being driven by both kinds of knowledge. Speakers have strong intuitions about the order of the two words: *salt and pepper* sounds far more natural than *pepper and salt*, even though there's nothing ungrammatical about the latter. For lower frequency binomials, Morgan and Levy (2016) demonstrated that these preferences are driven by abstract constraints such as the length of a word, the semantic content of a word, etc (Benor and Levy, 2006). However, that same study also showed that those constraints don't account for the ordering preferences of high-frequency binomials like *salt and pepper*. For those binomials, the preferences can't be predicted solely from the properties of the individual words, suggesting that the binomial is instead stored *holistically*. Binomial ordering preferences therefore show sensitivity to both abstract preferences and holistic storage depending on their frequency, which makes them a particularly useful case for studying the trade-off between the two.

1.2 Abstract Preferences in Language Models

Language models offer a way to study this trade-off that behavioral work cannot. Questions about the timecourse in which an expression stops being formed compositionally and starts being stored holistically is fundamentally one about experience, but a human speaker's history with any particular expression can only be estimated reasonably using corpus frequency. In a language model, that history is the training corpus itself, and it can be counted exactly. More importantly, in language models the representations are also directly inspectable, and the computation that produces them can be intervened on. Neither is possible for human learners.

The evidence for abstract preferences and holistic storage in language models has been more mixed than in the human behavioral literature. Models clearly generalize beyond what they have seen: Misra and Mahowald (2024) show that language models can produce the rare *a beautiful five days* construction even when it is withheld from training, generalizing to it from related constructions that are better attested, and Yao et al. (2025) find that dative alternation preferences are shaped by abstract constraints. Beyond text models, there is also evidence of abstractions for automatic speech recognition models as well (Pluth et al., 2026). At the same time, models retain a great deal of item-specific knowledge. McCoy et al. (2023) find that while model-generated text is often as structurally novel as human text, the same models also reproduce passages of over a thousand words verbatim from their training data. As such, it is clear that both abstract preferences and holistic storage are clearly at play within language models, however what remains unclear is how these types of knowledge trade off and when they emerge during training.

Perhaps most related to the present study, language models' ordering preferences for binomials appear not to be driven by abstract preferences at all. Houghton et al. (2025) examined whether language models showed the same abstract preferences as humans, and found that the models' ordering preferences were driven exclusively by frequency statistics, with no evidence of an effect of abstract preferences. However, that finding faces limitations along two main dimensions. First, the study can only claim to find no evidence that the models' ordering preferences were driven by *human* ordering preferences; it may very well be that their preferences are driven by abstract preferences that simply differ from those humans show. Second, the examination was conducted on model behavior, in the form of output probabilities, rather than over the representations that produce it. A model could in principle derive an ordering from abstract properties of the binomial and still produce output statistics that look frequency-driven.

1.3 Present Study

The present study thus asks whether binomial ordering preferences in language models are driven by abstract structure or if they are simply holistically stored. We make two changes to the design of Houghton et al. (2025). First, rather than testing model behavior against a fixed inventory of human constraints, we ask whether *any* systematic structure is recoverable at all, by training a probe to predict a model's ordering preference from its hidden representations and evaluating it on binomials that the probe was never trained on. Second, we separate the two components architecturally rather than statistically. Zeroing out cross-word attention

forces the model's preference to be purely compositional: a strictly additive function of the individual words evaluated separately. Interventions on attention have been used to trace factual recall and to isolate circuits (Geva et al., 2023; Wang et al., 2023), but to our knowledge this is the first use of one to separate holistic storage from abstract preferences. We apply this to 16 models across five families, spanning human-scale to trillion-token training corpora, and to training checkpoints for two families matched in parameters.

Our main contributions are:

- **Empirical evidence for both abstract preferences and frequency-driven holistic storage.** We show that a probe trained on LLM hidden-state representations for binomials can predict the ordering preferences for binomials it hasn't seen, however the binomial's relative frequency in the training data predicts a model's preference over and above anything those abstract preferences predict, and does so more strongly the more often the model encountered it.
- **An architectural intervention for isolating abstract preferences.** Zeroing out cross-word attention forces the representation into a strictly additive function of the two words, so the probe can only use what each word carries on its own.
- **Evidence that storage shifts lexical, not only phrasal, representations.** Zeroing out cross-word attention mitigates but does not eliminate the item-specific knowledge, indicating that repeated exposure to a phrase also changes the individual lexical representations themselves.

2 Methods

All three experiments apply the same procedure and differ only in which models[1] and checkpoints it is applied to. This section describes that procedure: the datasets, the probes, and the analyses.[2]

2.1 Datasets

In order to examine the ordering preferences for binomials, we created two binomial corpora: a novel set and an attested set. The **novel set** comprises approximately 340,000 binomials from the Wikitext corpus (Merity et al., 2016) selected to be absent from BabyLM's training corpus. The **attested corpus set** comprises approximately 49,000 binomials from the BabyLM corpus (Warstadt et al., 2023).

The novel set is genuinely novel only for the BabyLM models; for the large-scale models it consists of infrequent attested binomials rather than unseen ones. We retain the label *novel* throughout nonetheless.

Both sets were extracted automatically using a constituent parser requiring the two words to form a minimal coordinate phrase (Appendix B).

For each binomial and each model, we also calculate that model's ordering preference for the binomial, which is operationalized as the model's own log-probability ratio: $y_{\text{true}} = \log P(W_1 \text{ and } W_2) - \log P(W_2 \text{ and } W_1)$. Every binomial we probe occurs at least twice in Wikitext, and the two occurrences are used for different purposes: the probe's input representation is taken from one, and y_{true} is measured in the other. Measuring the preference in a sentence the probe never sees keeps it from recovering its own target (Appendix B).

For the developmental analysis in Experiment 3, we probe seven checkpoints from the BabyLM and Pythia models, the first being the untrained initialization. BabyLM checkpoints were sampled at approximately even intervals across training; Pythia checkpoints were instead chosen so that each model had seen roughly the same number of tokens as the corresponding BabyLM checkpoints. The trained checkpoints span roughly 20M to 2B effective training tokens.

2.2 Representation Extraction

For each model we extract final-layer hidden states, presenting each binomial in both orderings (W_1 and W_2; W_2 and W_1) within the same sentence frame. Each binomial is represented in two ways:

- **Binomial-level**: $[\bar{h}(W_1 \text{ and } W_2); \bar{h}(W_2 \text{ and } W_1)]$, where $\bar{h}$ denotes the mean of token representations across all tokens in the binomial in each ordering.
- **Word-level**: $[h(W_1); h(W_2)]$, where $h(W_k)$ is the hidden state of word W_k taken from the sentence in which it appears in first position.

[1] Appendix A lists all sixteen models with their parameter counts and training-token estimates.

[2] All code and analyses in this paper can be found here: https://anonymous.4open.science/r/binom-representation-pls-6EA0/.

2.3 Attention Conditions

We additionally probe representations under two attention conditions:

- **Full context**: standard forward pass with full cross-token attention. Each word's representation reflects contextual influences from all other (previous) tokens in the sentence.
- **Attention-zeroed**: the same forward pass, except that we zero out attention from the second word and the conjunction to the first word, at every layer (Appendix C).

This gives three representation types per model: binomial-level under each attention condition, and full-context word-level.

Because y_{true} is a difference of log-probabilities over the two orderings, and because causal masking already prevents W_1 from attending to W_2, blocking attention from W_2 and the conjunction to W_1 leaves the model blind to the other word. The resulting preference is necessarily of the form $g(W_1) - g(W_2)$: an additive function of the two words evaluated separately, with no interaction term. The attention-zeroed model therefore reports the preference it would hold given only word-level information.

2.4 Probes

For each representation type, we train a two-layer MLP regressor (hidden dimension 128, ReLU activation) to predict binomial ordering preferences from the binomial representations. The probe takes the concatenated representations for a binomial and outputs a single continuous value $\hat{y}$, trained under mean-squared-error loss against y_{true}, with antisymmetric augmentation so that a swapped ordering carries the negated label. Optimizer settings, early stopping, and the augmentation rate are given in Appendix C. We report cross-validated R^2 (the squared Pearson correlation between $\hat{y}$ and y_{true} on held-out folds), averaged across 10 folds.[3]

We evaluate generalization under two 10-fold cross-validation conditions:

- **Binomial-held-out**: binomials are held out as units; each binomial in the test set is excluded from the training set, however overlapping words are allowed (e.g., *radios and computers* is not excluded from the training set even if *radios and televisions* is in the test set).
- **Word-held-out**: binomials with overlapping words are excluded from the training set. For example, if *radios and televisions* is in the test set, then *radios and computers* is excluded from the training set.

Both splits are applied to the novel set; the attested set is not cross-validated, and is instead used to evaluate the effects of binomial frequency on the probe's predictions. Two senses of novelty are in play here, and they are nested. The novel set is novel with respect to the *language model's* training corpus, in that the pairing does not occur there; the two splits then control what the *probe* has seen within that set. An item in the word-held-out test set is thus new to the probe in both its pairing and its two words, while those same words are ordinary vocabulary that the language model encountered many times. We report fold standard deviations throughout (Appendix E).

Finally, following Hewitt and Liang (2019), we train a matched control probe on randomly permuted labels; near-zero control R^2 indicates the probe is recovering linguistic structure rather than overfitting to noise.

2.5 Analyses

We report two measurements. The first is the probe's cross-validated R^2 under the two splits described prevously, of which word-held-out is the stricter (since binomial-held-out R^2 may be inflated by general lexical preferences).

The second measurement evaluates the probe on the attested set. Since the probe has seen only binomials the model has not encountered (or encountered rarely in the case of the larger models), its prediction $\hat{y}$ for an attested binomial estimates what abstract knowledge alone would predict, and cannot reflect the probe (or the LLM) having memorized that binomial. In order to analyze the extent to which the model's ordering preference within the attested set is driven by abstract vs holistic knowledge, we implement a Bayesian regression model using *brms* (Bürkner, 2017) with default priors. Our dependent variable is y_{true}, the model's ordering preference. The probe's prediction, $\hat{y}$, stands in for abstract preferences. Relative frequency, RelFreq, is the centered proportion of occurrences in alphabetical ordering: 0.5 means the binomial always occurs in alphabetical ordering

[3] The results below do not depend on the MLP's capacity. A ridge regression on the same data, using the same folds and a single tuned hyperparameter, recovers most of the MLP's performance. Full comparison in Appendix D.

in the corpus, 0 means it occurs evenly in both orderings, and -0.5 means it always occurs in non-alphabetical ordering. Letting log(Frequency) be log overall frequency (overall occurrences in the model's corpus, logged[4]), our regression equation is:

$$\begin{aligned} y_{\text{true}} \sim \hat{y} * \log(\text{Frequency}) \\ + \text{RelFreq} * \log(\text{Frequency}) \\ \log(\sigma) \sim \log(\text{Frequency}) \end{aligned} \quad (1)$$

y_{true} and $\hat{y}$ are centered and scaled within each fit; log(Frequency) is centered but not scaled; RelFreq is left on its original scale, which already runs from -0.5 to $+0.5$ (Appendix G).

The residual scale is modeled rather than assumed constant. The spread of y_{true} varies with frequency, and an unmodeled trend there would inflate the slopes at one end of the range.

3 Experiment 1: BabyLM Representations

In this first experiment, we probe three BabyLMs (125M, 350M, and 1.3B parameters)[5] at their final training checkpoint. Because these models were trained on an amount of data comparable to humans, and because the novel set excludes binomials in their training corpus, they give the cleanest test of whether a language model encodes abstract binomial ordering preferences.

3.1 Results

3.1.1 Probe Performance

Results are shown in Figure 1; full numerical results are in Appendix H (Table 5). The control probe R^2 values were consistently near zero across all conditions and all three models (reported alongside each R^2 in Table 5), confirming that the probe doesn't recover spurious signal from noise.

For the binomial-level conditions, binomial-held-out R^2 ranges from 0.04 to 0.13; word-held-out R^2 is consistently lower, ranging from 0.04 to 0.10. The drop reflects a lexical generalization cost: some of the preference signal is tied to word identity rather than being fully abstract. Probe accuracy falls monotonically with model size under both attention conditions, from 0.13 at 125M to 0.07 at 1.3B under full context, and the full-context condition yields higher binomial-held-out R^2 than attention-zeroed in all three models.

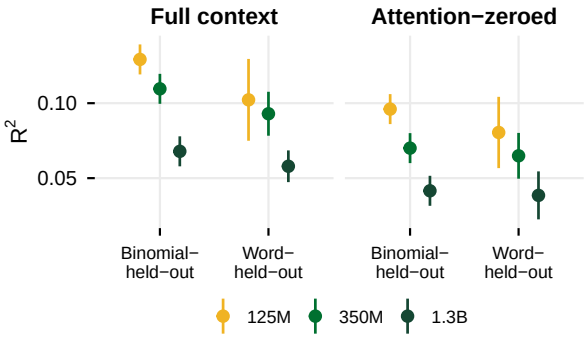


Figure 1: Cross-validated probe R^2 at the final hidden layer for the BabyLM models, binomial level, by attention condition. Bars = ±1 SD across 10 folds.

3.1.2 Abstract and Stored Contributions

Results are shown in Figure 2. Both abstract preferences and relative corpus frequency predict the model's ordering preference, and both do so credibly in all three models (Appendix G). The effect of relative frequency is stronger for high-frequency binomials, while the effect of abstract preferences is largely flat across frequencies, declining credibly only in the 350M.

Zeroing out attention reduces both the effects of abstract preferences and relative frequency for the BabyLMs, and makes the interaction between abstract preferences and overall frequency credibly negative in all three, suggesting that abstract preferences are a slightly *worse* predictor of ordering preferences for high-frequency binomials when attention is zeroed.

4 Experiment 2: Cross-Family Final-Layer Representations

Experiment 1 demonstrates that binomial ordering preferences within the BabyLMs are driven by both abstract preferences and holistic storage. Those models are small in both parameters and training data, so Experiment 2 applies the same probe to 13 further models across four large-scale families: Pythia, GPT-2, OLMo, and Llama, with the BabyLM results carried over for comparison (Appendix A).

[4]Frequency is each binomial's exposure in the model's own training corpus: exact counts for BabyLM (Warstadt et al., 2023) and Pythia, and infini-gram Pile counts (Gao et al., 2020; Liu et al., 2024) standing in as a proxy for the rest. Appendix F gives the distributions.

[5]The BabyLM models used here are OPT-based models trained for 20 epochs on the BabyLM v3 corpus (Houghton et al., 2026; Warstadt et al., 2023).

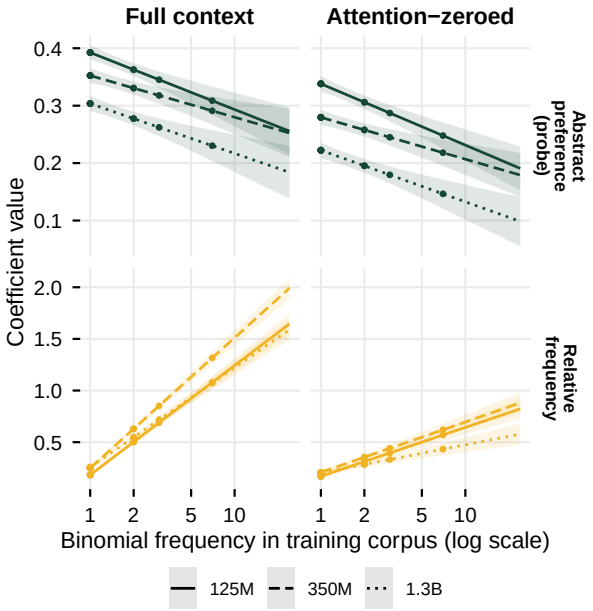


Figure 2: How strongly each source contributes to the BabyLM models' ordering preference, by binomial frequency. Values are slopes from the fitted posteriors of Equation 1. Rows are the two sources, each on its own scale. Bands are 95% credible intervals.

4.1 Results

4.1.1 Probe Performance

Figure 3 shows both cross-validation splits for all 16 models; full numbers, control probes, and the word-level replication are in Appendix I. Control R^2 was near zero throughout, never exceeding 0.001. All large-scale models substantially outperform BabyLM: full-context binomial-held-out R^2 spans 0.16 to 0.21 against 0.07–0.13, and word-held-out R^2 is lower than binomial-held-out in every one of the 32 model-by-condition cells. Attention-zeroing lowers binomial-held-out R^2 in all 16 models, by between 0.03 and 0.10, with the largest reductions in the 7B and 8B models and in Pythia-1B, and the smallest in the BabyLMs. Scale effects within families differ in direction rather than being uniformly non-monotonic: R^2 rises with parameters in Pythia (0.16 to 0.20) and OLMo (0.19 to 0.21), is flat across GPT-2 (0.18 at every size), and falls in BabyLM (0.13 to 0.07).

4.1.2 Abstract and Stored Contributions

Results are shown in Figure 4. Similar to the BabyLMs, there is an effect of both abstract preferences and relative frequency for all models. Further, the effect of relative frequency is stronger for frequent binomials compared to infrequent ones. Unlike the BabyLMs, however, the effect of abstract preferences *decreases* as a function of frequency. In other words, the more the models encounter a binomial, the more their ordering preferences are governed by that binomial's frequency and the less by abstract preferences.

Zeroing out attention mitigates the effect of relative frequency substantially for both the main-effect and the interaction with overall frequency, however both remain credible. Further, the effect of abstract preferences *increases* in nine of the thirteen large-scale models; of the four exceptions, all of them among the largest, only Llama-3-8B decreases credibly, the other three falling within their credible intervals. The interaction effect grows more negative in all but one. In other words, the effect of abstract preferences is generally stronger with attention zeroed out, but this increase is weaker for high-frequency binomials.

Notably, the effect of zeroing out attention on abstract preferences differs between the BabyLMs and most of the other models. For the BabyLMs, zeroing out attention weakens the effect of both abstract preferences and relative frequency; for most of the large-scale models, it weakens the effect of relative frequency but *strengthens* the effect of abstract preferences. The reduction in the effect of relative frequency, by contrast, is of a similar size in the two groups, ranging from 33% to 42% in the BabyLMs against 30% to 47% in the others. We return to possible explanations for this contrast in the Discussion and Conclusion section.

5 Experiment 3: Developmental Trajectories

Experiments 1 and 2 establish that language models encode binomial preferences at the end of training, but leave open when those representations emerge, and whether the trade-off between abstract preferences and holistic storage is present early or accumulates with exposure.

The probe, inputs, attention conditions, and cross-validation splits are as described previously. We refit Equation 1 at each saved checkpoint of the three BabyLM models and the three Pythia models, in both attention conditions.

5.1 Results

Probe trajectories are shown in Figure 5 and the two sources' contributions in Figure 6. Full numerical and word-level results are in Appendix J. Binomial-held-out R^2 begins at essentially zero in every

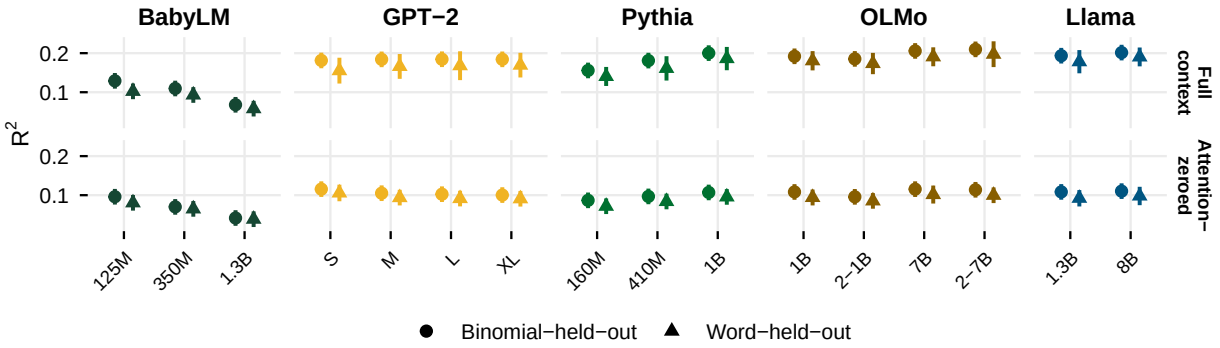


Figure 3: Cross-validated R^2 for all 16 models, by attention condition and cross-validation split. Circles = binomial-held-out; triangles = word-held-out. Bars = 95% CI across 10 folds. Models ordered by parameter count within each family and colored by family.

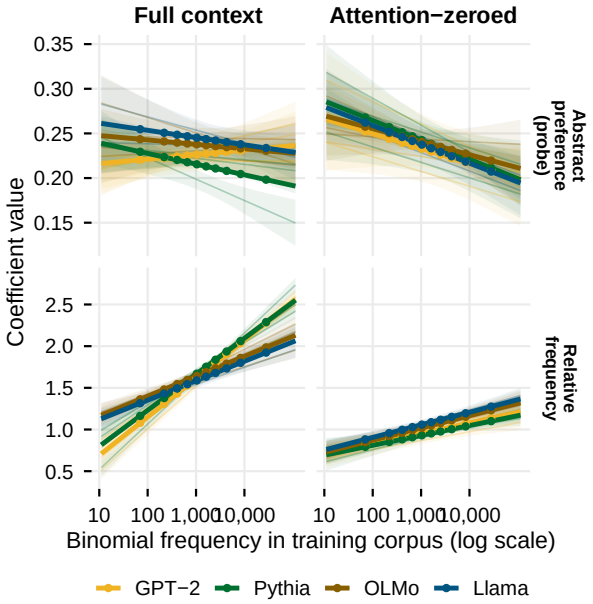


Figure 4: How strongly each source contributes to the ordering preference, by binomial frequency, for the 13 large-scale models. Values are simple slopes from the fitted posteriors of Equation 1. Rows are the two sources, each on its own scale. Faint lines are individual models with their 95% credible intervals, bold lines the mean within each family.

model and rises with training. For the BabyLMs it is 0.00 at the earliest checkpoint (~20M tokens), climbs to 0.13–0.14 by mid-training, and stands at 0.10–0.14 at the end. Pythia follows the same course, rising from 0.00–0.01 at 33.6M tokens to 0.15–0.19 at 2097M tokens. Full context exceeds attention-zeroing at the final checkpoint in all three BabyLMs, while in Pythia the two conditions end close together.

The effect of relative frequency is small across the first three checkpoints of every model, never exceeding 0.29 (compared to a maximum value of 1.03) across training. From the fourth checkpoint it is credibly positive in every model and climbs thereafter to 0.353, 0.447 and 0.512 in the three BabyLM models, and to 0.995, 0.914 and 1.031 in the three Pythia models. This holds in both attention conditions, so what grows is not solely the binomial-specific representation. At step 0, before the models have seen any data, the effect is indistinguishable from zero in the three BabyLM models, but credibly positive in the three Pythia models (0.11–0.15); this Pythia floor is an artifact of subword tokenization rather than any ordering preference in the untrained networks.[6]

At the fourth checkpoint, the first at which relative frequency is credibly positive in every model, attention-zeroing removes none of the effect. In other words, at the point where storage is first measurable, it survives an intervention that removes the phrasal representation, so what has shifted at that point is the lexical representations. The phrasal representations shift afterwards. This is demonstrated from the amount of the relative-frequency effect that is removed from zeroing out cross attention. After the fourth checkpoint, the amount that the the relative-frequency effect removed by zeroing out attention rises from nothing at the fourth checkpoint to 13–21% by the last in the three Pythia models and 27–42% in the three BabyLM mod-

[6]This is a tokenization artifact: for some binomials the two orderings segment into different numbers of tokens, and at initialization each token costs roughly the log of the vocabulary size, so the difference between them is large for reasons that have nothing to do with ordering preferences. Where the preceding context ends in a quotation mark rather than a space, for instance, the first word loses its leading space, and *"bread and butter"* is three tokens (`bread`, `and`, `butter`) while *"butter and bread"* is four (`but`, `ter`, `and`, `bread`). Roughly 2% of items are affected. Excluding them eliminates the effect of relative frequency at the step 0 checkpoint but leaves it unchanged in the fully trained models; Appendix K gives the full comparison.

els, and reaches 30–47% in the fully trained large-scale models. The interaction effect between relative frequency and overall frequency follows the same course, reaching 0.191, 0.288 and 0.351 in the BabyLMs and 0.121, 0.104 and 0.130 in the Pythia models.

The effect of abstract preferences follows its own course. It rises across training in both families and in both conditions, from near zero at the earliest checkpoints to 0.374, 0.364 and 0.324 in the three BabyLM models and 0.310, 0.326 and 0.327 in the three Pythia models, with most cells reaching their maximum a checkpoint or two before the end. A small effect is nonetheless present at step 0 under full context, before the models have seen any data, though not under attention-zeroing.[7] Its interaction with overall frequency is credibly negative under attention-zeroing in 17 of the 36 model-checkpoint fits (six checkpoints for each of the six models).

Both sources are coefficients in the same regression, fit to the same items, so we can ask when each becomes measurable on a single scale. We use two criteria. The first is detectability: the earliest checkpoint whose posterior sits credibly above the same coefficient's value at step 0, in the same model and condition, and stays there. Taking step 0 rather than zero as the baseline keeps an untrained network from registering, which matters here because both coefficients are already non-zero before training for reasons that have nothing to do with learning. The second is trajectory: the earliest checkpoint at which a coefficient has covered three quarters of its own change from step 0 to its final value. The two fail in opposite directions, the first favoring whichever measure is more sensitive and the second ignoring sensitivity altogether.

Abstract preferences come first on both counts. They are detectable from 20 to 134M tokens against 118 to 315M for relative frequency, and cover three quarters of their change by 118 to 1074M against 1074 to 2097M. The interaction between relative frequency and overall frequency is last on both, detectable from 298 to 1074M tokens. Per-model values are in Appendix L. We read this as soft evidence about ordering rather than as an estimate of when either source becomes available; the two measures are not on equal footing, for reasons we return to in the Limitations.

[7] Novel binomials are novel as pairings, not as words: 97% of attested binomials have both of their words somewhere in the probe's training set. A randomly initialized network already assigns each word an arbitrary but consistent bias, so the probe can learn those biases and carry them over to attested pairs built from the same words, and the effect scales with that overlap, from $+0.12$ where the probe had seen both words to a non-credible estimate where it had seen neither. It accounts for roughly 1% of the variance at step 0 against 17% in the trained model.

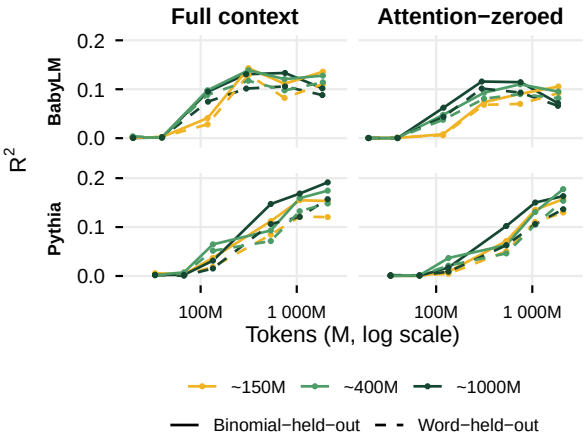


Figure 5: Cross-validated probe R^2 across BabyLM and Pythia training checkpoints, by attention condition and cross-validation split. Control $R^2 \approx 0$.

6 Discussion and Conclusion

The present study examined whether language models' binomial ordering preferences are driven by abstract structure, and if so, how that structure emerges throughout training. Across 16 models from five families, we found evidence for ordering preferences being driven by both abstract and item-specific knowledge. Preferences generalize to binomials that are absent from the language model's training corpus, and continue to do so when the probe is additionally denied both of a binomial's words during its own training, demonstrating abstract structure that is shared across independently trained models. At the same time, a binomial's relative frequency contributes over and above abstract preferences, and contributes more as the binomial becomes more frequent. These two sources also behave differently over training: abstract preferences become measurable earlier than the item-specific contribution does (though note that only being able to measure something later does not mean the information wasn't present earlier). Finally, zeroing out cross-word attention removes only part of the relative-frequency contribution, suggesting that storage shifts the lexical representations in addition the phrasal one.

The evidence for abstract preferences is particularly strong because it does not depend on generalization across familiar words alone. The probe recovers a model's preference for binomials that are

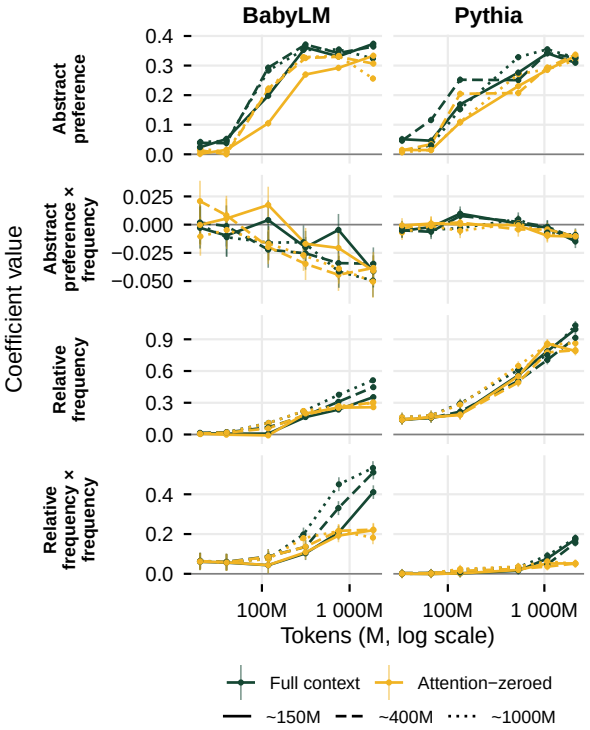


Figure 6: All four terms of Equation 1 across training, with 95% credible intervals. Rows are the two sources and their interactions with overall frequency; color marks the attention condition and linetype the model size.

absent from the language model's own training corpus, and it does so even when both of the binomial's words are additionally withheld from the probe's training folds. The two exclusions operate at different levels: the pairing is new to the language model, and the words are new to the probe. The language model has of course seen those words many times. A ridge probe with a single tuned hyperparameter recovers the same pattern, showing that the result does not depend on the MLP's capacity (Appendix D). The answer to the question left open by Houghton et al. (2025) is therefore that models do have abstract preferences, but ones that modeling with human constraints seems to have failed to find. Further evidence that these preferences are systematic comes without the probe at all: independently trained models converge on the same preferences for the same novel binomials ($r = 0.50$, against a shuffled baseline of $r \approx 0.00$). This agreement holds even among the three BabyLM models, which share a training corpus but differ in capacity, and is stronger within model families than between families, though it remains high between families as well. Because all sixteen models received the same sentence frame, the latter pattern cannot be attributed to the frame itself.

Abstract structure, however, does not fully determine a model's preference for a particular binomial. A binomial's relative frequency predicts the model's preference over and above what the abstract structure predicts, and this item-specific contribution increases with frequency. Unlike the abstract structure, this information is specific to the particular binomial. The result therefore reveals a frequency-dependent division between generalizable structure and experience with individual items, closely paralleling the trade-off between abstract preferences and item-specific experience observed by Morgan and Levy (2016) in humans.

Abstract preferences come first. Probe R^2 begins at essentially zero and rises over training, and in Equation 1 the abstract component becomes measurable earlier than the item-specific one in every model, on both of the criteria we apply to them. The item-specific contribution follows a different trajectory: it is negligible through the first three checkpoints of every model, becomes credibly positive everywhere from the fourth, and then climbs monotonically under full context without leveling off. This is not conclusive evidence against item-specific knowledge being learned early, however, since the measurement relies on a frequency-difference in representations to detect holistic storage. The absence of a frequency difference, consequently, does not necessarily indicate lack of storage.

Finally, the attention intervention provides further insight into how this item-specific information is represented. Zeroing out cross-word attention mitigates but does not eliminate the contribution of relative frequency: at the first checkpoint where a binomial's relative frequency predicts the model's ordering preference at all, attention-zeroing removes none of that prediction; the share removed then increases over training. This pattern suggests that item-specific information is not exclusively stored as a holistic representation of the binomial. Instead, it seems possible that repeated exposure to a binomial gradually shifts the representations of its individual words, so that a word encountered overwhelmingly in first position comes to encode something of that tendency on its own. Such information remains item-specific in that it concerns particular words and their learned positional tendencies rather than a general constraint, but it no longer requires the model to represent the specific binomial context. On this interpretation, entrenchment shifts the

lexical representations before it shifts the phrasal one. This interpretation parallels findings in the human behavioral literature showing that words experienced frequently in specific contexts can come to reflect that change even when encountered outside those contexts (Kapatsinski, 2026).

Overall, the findings bridge a longstanding gap between artificial language models and usage-based accounts of language. Such accounts have long proposed that frequent expressions become increasingly entrenched, with behavioral work on binomials showing that human ordering preferences reflect both abstract constraints and holistic storage. The present results show a similar division emerging in a system with no innate linguistic preferences: trained only to predict text, the models learn both abstract representations and holistically stored representations. The results therefore suggest that the interaction they don't need separate mechanisms explicitly built into a learner, but rather they can emerge from exposure itself within a distributed representational system.

Limitations

The novel binomials were selected to be absent from BabyLM's training corpus, so they are genuinely novel only for the three BabyLM models; indexed against Dolma (Soldaini et al., 2024), a trillion-token corpus, roughly 98.4% occur at non-zero frequency. For the thirteen large-scale models, then, the probe is trained on infrequent attested binomials, not unseen ones. This is why Experiment 1 anchors the interpretation. Corpus frequency is likewise an exact measure of what the model saw only for BabyLM and Pythia; for GPT-2, OLMo, and Llama, Pile counts stand in as a proxy for web-scale exposure.

The two sources in Equation 1 are also not measured on equal footing, which is why we treat the ordering in Experiment 3 as soft evidence. $\hat{y}$ is a fitted readout of the model at that checkpoint, so it tracks whatever the model currently encodes. Relative frequency is a fixed corpus statistic measured with error, and measurement error in a predictor pulls its coefficient toward zero. It is also computed over the whole training corpus rather than over what the model has seen by that checkpoint, so early on it describes text the model has not encountered yet. Both asymmetries push storage later and abstract preferences earlier, which is the direction our ordering runs. The BabyLM checkpoints make the problem concrete: they sit at fixed epoch fractions, and relative frequency becomes detectable at the first checkpoint past one complete pass over the corpus in all three models. That is what we would expect if storage requires repeated exposure, and equally what we would expect if the predictor only starts describing the model's experience after one pass. Our data cannot separate the two.

Additionally, the results are limited to English and to a single construction.

Finally, the intervention separates additive from non-additive information, not abstract from holistic information directly. The two line up only if abstract constraints can be stated over each word on its own. Most of the constraints in Morgan and Levy (2016) can be: relative word length or frequency compares two quantities that each word carries by itself, so an additive representation preserves them. A constraint over the juncture between the two words is different. Avoiding a stress clash across the boundary, for instance, is abstract and generalizable, but the mask would likely remove it and we would count it as holistic storage. The dissociation between the two effects suggests this isn't what's driving our results, since losing information indiscriminately would have weakened both effects rather than only one. Even so, the design doesn't rule it out.

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A Model Details

Table 1 lists all 16 models used across experiments, organized by family, with approximate parameter counts and training token counts.

Family	Model	Parameters	Tokens
BabyLM	BabyLM-125M	125M	150M
	BabyLM-350M	350M	150M
	BabyLM-1.3B	1.3B	150M
GPT-2	GPT-2	117M	~8B
	GPT-2-medium	345M	~8B
	GPT-2-large	762M	~8B
	GPT-2-XL	1.5B	~8B
Pythia	Pythia-160M	160M	~300B
	Pythia-410M	410M	~300B
	Pythia-1B	1B	~300B
OLMo	OLMo-1B	1B	3T
	OLMo-7B	7B	2.5T
	OLMo-2-1B	1B	~4T
	OLMo-2-7B	7B	~4T
Llama	Llama-1.3B	1.2B	~9T
	Llama-3-8B	8B	~15T

Table 1: Models used across experiments. Training token counts for full-scale models are approximate; GPT-2 counts are estimated from the reported 40GB WebText corpus. BabyLM models train for 20 epochs over 150M tokens (3B effective token exposures).

B Binomial Extraction Criteria

Binomials ("W1 and W2") were first selected using spaCy (Honnibal et al., 2020) part-of-speech tags, retaining only cases where both W_1 and W_2 are open-class words (nouns, verbs, adjectives, or adverbs). Each candidate was then parsed with the Berkeley Neural Parser (Kitaev et al., 2019; Kitaev and Klein, 2018) to confirm that the two words form a minimal coordinate phrase. The inclusion criterion was that the two words be co-constituents under the same phrasal node (NP, VP, ADJP, ADVP, PP, or equivalent), with that node spanning exactly the three tokens [W_1, and, W_2] and nothing else. This ensures the extracted items are genuine coordinate binomials rather than incidental sequences where "and" links clauses or longer phrases. Novel binomials were extracted from Wikitext by the identical procedure, with the added requirements that both words appear in BabyLM's vocabulary and that the binomial is absent from the attested set.

Because our extraction criteria requires W1 and W2 to form a phrase-level constituent spanning exactly W1, "and," and W2, binomials that do not satisfy this constituency criterion are excluded. We do not additionally require the two conjuncts to receive the same POS tag, because spaCy's POS tagger can misclassify participial forms in coordinate contexts, for example tagging "mortified" as VERB rather than ADJ in "disgusted and mortified." Requiring POS-tag agreement would therefore risk excluding valid binomials because of POS-tagging errors.

Two further restrictions apply to the items we probe. First, each binomial must occur at least twice in Wikitext with at least ten characters of preceding text, since we measure the representations in one occurrence and the ordering preference in the other, and a binomial at the start of a sentence has no left context to distinguish it from any other such item. Second, reduplicatives ("again and again") are excluded throughout: their two orderings are the same string, so the ordering preference is identically zero and carries no information. Of the 340,042 novel binomials the extraction yields, 216,523 satisfy both conditions, and it is these that the probes and the frequency regression use. Retention is higher for frequent binomials than for rare ones, but it is unrelated to which ordering a binomial prefers: among items attested once in the training corpus, the mean relative frequency of those retained is -0.004 against -0.000 for those dropped.

C Implementation Details

Probe training. The probe is optimized with Adam (learning rate 10^{-3}, weight decay 10^{-4}) under mean-squared-error loss, with early stopping (patience 5) against a 10% held-out validation split. Antisymmetric augmentation is applied with probability 0.5: for each binomial the flipped version, with swapped ordering and negated label, is added to the training batch. The input dimensionality follows from the representation type.

Attention-zeroing. The zero weight is imposed before the softmax, by adding a large negative constant to the corresponding attention logits, so that the remaining positions renormalize to sum to one.

Recomputing the target. Because y_{true} is the model's own log-probability ratio, it is recomputed from whichever forward pass produced the representations being probed. The two full-context representation types therefore share a target, while the attention-zeroed condition has its own: blocking cross-word attention changes the model's output probabilities as well as its hidden states.

D Linear Probe Replication

The MLP probe has many free choices: hidden dimension, depth, activation, learning rate, weight decay, patience, validation fraction, augmentation probability, batch size, and initialization. A skeptical reading is that some of them were tuned toward the reported result. We therefore repeat the analysis with a probe that has one.

The linear probe is a ridge regression on the same features, fit on the same cross-validation folds. Its only free variable is the ridge penalty, selected per fold from $\{10^0, \ldots, 10^5\}$ on an inner split of that fold's training data. That inner split is the same size as the MLP's early-stopping split, so both probes receive an identical budget of held-out data for hyperparameter selection.

Across 16 model-checkpoint cells and 64 comparisons, the two probes agree closely ($r = 0.985$ between their R^2 values). On the word-held-out split the linear probe recovers 103% of the MLP's R^2 on average (0.125 vs. 0.122).

Model	Condition	Split	Linear	MLP	Gap
BabyLM-125M	Attention-zeroed	B	0.093	0.096	+0.003
		W	0.086	0.080	-0.006
	Full context	B	0.114	0.129	+0.016
		W	0.104	0.102	-0.001
BabyLM-350M	Attention-zeroed	B	0.072	0.070	-0.002
		W	0.069	0.065	-0.004
	Full context	B	0.102	0.110	+0.008
		W	0.099	0.093	-0.006
BabyLM-1.3B	Attention-zeroed	B	0.055	0.042	-0.013
		W	0.050	0.038	-0.011
	Full context	B	0.081	0.068	-0.013
		W	0.078	0.058	-0.020
GPT-2	Attention-zeroed	B	0.109	0.115	+0.007
		W	0.100	0.106	+0.006
	Full context	B	0.167	0.181	+0.014
		W	0.150	0.155	+0.005
GPT-2-M	Attention-zeroed	B	0.103	0.105	+0.002
		W	0.093	0.094	+0.001
	Full context	B	0.177	0.184	+0.007
		W	0.163	0.166	+0.003
GPT-2-L	Attention-zeroed	B	0.098	0.103	+0.004
		W	0.090	0.091	+0.001
	Full context	B	0.181	0.184	+0.003
		W	0.170	0.168	-0.002
GPT-2-XL	Attention-zeroed	B	0.100	0.100	-0.000
		W	0.091	0.091	-0.000
	Full context	B	0.186	0.184	-0.001
		W	0.172	0.169	-0.002
Pythia-160M	Attention-zeroed	B	0.077	0.087	+0.010
		W	0.066	0.072	+0.006
	Full context	B	0.141	0.156	+0.015
		W	0.135	0.140	+0.005
Pythia-410M	Attention-zeroed	B	0.090	0.097	+0.006
		W	0.076	0.084	+0.008
	Full context	B	0.167	0.181	+0.014
		W	0.157	0.161	+0.004
Pythia-1B	Attention-zeroed	B	0.109	0.107	-0.002
		W	0.095	0.096	+0.000
	Full context	B	0.200	0.200	+0.000
		W	0.187	0.186	-0.000
OLMo-1B	Attention-zeroed	B	0.106	0.108	+0.002
		W	0.093	0.094	+0.001
	Full context	B	0.191	0.192	+0.001
		W	0.183	0.181	-0.003

(continued)

Model	Condition	Split	Linear	MLP	Gap
OLMo-7B	Attention-zeroed	B	0.127	0.115	-0.012
		W	0.112	0.102	-0.010
	Full context	B	0.219	0.206	-0.013
		W	0.209	0.191	-0.018
OLMo-2-1B	Attention-zeroed	B	0.097	0.096	-0.001
		W	0.085	0.085	+0.000
	Full context	B	0.191	0.185	-0.005
		W	0.178	0.174	-0.005
OLMo-2-7B	Attention-zeroed	B	0.120	0.114	-0.006
		W	0.107	0.100	-0.007
	Full context	B	0.225	0.210	-0.016
		W	0.214	0.197	-0.016
Llama-1.3B	Attention-zeroed	B	0.106	0.108	+0.002
		W	0.092	0.092	+0.000
	Full context	B	0.196	0.193	-0.002
		W	0.179	0.178	-0.001
Llama-3-8B	Attention-zeroed	B	0.118	0.110	-0.008
		W	0.107	0.098	-0.010
	Full context	B	0.214	0.201	-0.013
		W	0.206	0.190	-0.016

Table 2: Linear (ridge) versus MLP probe R^2 at the final hidden layer, binomial-level input mode. Both probes use identical cross-validation folds. B = binomial-held-out; W = word-held-out. Gap = MLP minus linear.

The two probes agree closely, and the generalization pattern is the same under both. The reported results therefore do not depend on the MLP's extra capacity.

E Cross-Validation Split Sizes

Cross-validation is carried out on the novel set only. The attested set is never cross-validated: it is used solely as the transfer target described in Section 2.5. Both splits use ten folds, and the same folds are reused across models, attention conditions, and probe architectures so that all reported R^2 values are directly comparable.

Under the binomial-held-out split, binomials are assigned to folds directly, so each fold trains on nine tenths of the set and tests on the remaining tenth. Under the word-held-out split it is the *vocabulary* that is partitioned into ten folds. A binomial enters the test set for fold k only if both of its words are in fold k, and enters the training set only if neither word is. Binomials with one word in fold k and one word outside it are used in neither, and are discarded for that fold. Since a binomial needs both words drawn from the same tenth of the vocabulary to be testable, the word-held-out test set is roughly a hundredth of the set rather than a tenth. Table 3 gives the resulting sizes.

Split	Train / fold	Test / fold	Discarded / fold
Binomial-held-out	194,870–194,871	21,652–21,653	0
Word-held-out	172,411–176,774	1,961–2,548	37,747–41,564 (18.0%)

Table 3: Cross-validation fold sizes for the novel set (216,523 binomials over 36,209 unique words). Ranges are across the 10 folds. Discarded items under word-held-out are binomials with exactly one word in the held-out vocabulary fold.

F Corpus Frequency Distributions

The figure below compares the frequency distributions of the 48,965 corpus binomials in the BabyLM training corpus (150M tokens, 20 epochs) and the Pile, illustrating the difference in lexical coverage between the two training regimes. BabyLM's distribution is dominated by singleton binomials (71.2% appear exactly once), while Pile frequencies span several orders of magnitude.

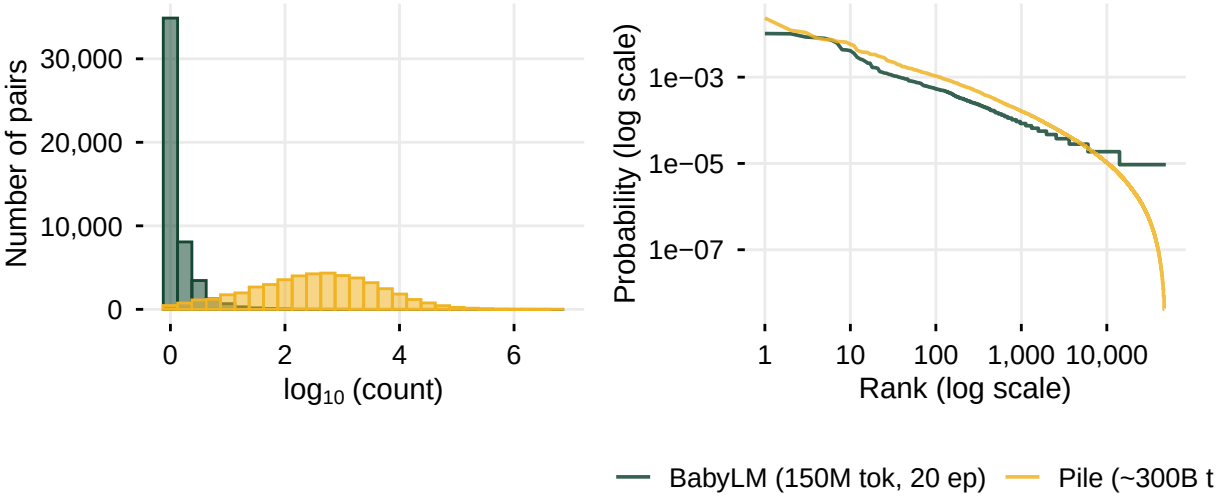


Figure 7: Frequency distributions for the 48,965 corpus binomials in the BabyLM training corpus (150M tokens, 20 epochs) and the Pile. Left: histogram of $\log_{10}$ raw counts. Right: rank-frequency plot treating counts as a probability distribution (probability = count / $\sum$ counts); both axes log-scaled. BabyLM's distribution is heavily concentrated at a single occurrence per pair (71.2% of pairs appear exactly once), whereas Pile frequencies span several orders of magnitude with a median of 365 occurrences per pair.

The BabyLM training corpus is a 150M-token collection of child-directed and informal text (CHILDES, BNC Spoken, children's books, OpenSubtitles, Wikipedia), trained for 20 epochs. The Pile is a ~300B-token web-scraped corpus used to train the Pythia family. For each of the 48,965 corpus binomials, we extracted 3-gram counts (i.e., the frequency of "word_1 and word_2" and its reverse) from each corpus via infini-gram (Liu et al., 2024). As Figure 7 shows, the BabyLM corpus produces a highly skewed frequency distribution: 71.2% of binomials appear exactly once, 87.7% appear at most twice, and 98.2% appear at most ten times, with a maximum of 1,095 occurrences. The Pile distribution is far richer: the median binomial appears 365 times, the 25th percentile is 62 occurrences, and the 99th percentile exceeds 77,000.

In absolute training exposure terms (accounting for BabyLM's 20 epochs), the most frequent BabyLM binomial that enters our analyses, *back and forth*, is seen ~21,600 times total across all training steps, while the same binomial is seen over 1.5 million times in a single pass of the Pile. Even the median Pile binomial (365 occurrences per pass) substantially exceeds BabyLM's total exposure for a typical singleton binomial ($1 \times 20 = 20$ exposures).

G Frequency-Dependent Preference Regression

Full results for Equation 1: abstract preferences $\hat{y}$ (y_pred), the binomial's relative frequency in the corpus (rel_freq), overall binomial frequency (log_freq), the two moderations, and the residual-scale term.

We first justify the z-scoring of y_true and y_pred and *not* z-scoring rel_freq and log_freq. y_true and y_pred are properties of the model: both are log-probability ratios, so a unit is nominally the same everywhere, but the spread of that quantity reflects how sharp a given model's output distribution is rather than how much it knows about binomials. Across the models here the standard deviation of y_true ranges from 2.79 to 4.48, and it bears almost no relation to probe R^2 ($r = 0.15$ across the sixteen models, and it runs in the opposite direction within BabyLM), so a raw coefficient would partly measure output calibration. We therefore z-score both within each fit, which puts every coefficient in units of that model's own preference spread.

rel_freq and log_freq, by contrast, are properties of the training corpus: a proportion of occurrences and a count of them mean the same thing however they are measured, and dividing either by a corpus-specific standard deviation would replace a shared unit with a local one. They are therefore left on their own scales: rel_freq is a proportion running from -0.5 to $+0.5$ and log_freq is centered but not scaled, so both mean the same thing in every corpus.

Model	Condition	y_pred	rel_freq	y_pred × log_freq	rel_freq × log_freq
BabyLM-125M	Full context	0.374 [0.363, 0.384]	0.386 [0.363, 0.409]	-0.043 [-0.058, -0.028]	0.460 [0.425, 0.495]
	Attention-zeroed	0.318 [0.306, 0.329]	0.260 [0.237, 0.284]	-0.046 [-0.060, -0.033]	0.205 [0.172, 0.240]
BabyLM-350M	Full context	0.339 [0.328, 0.350]	0.490 [0.467, 0.514]	-0.032 [-0.046, -0.016]	0.548 [0.513, 0.583]
	Attention-zeroed	0.266 [0.255, 0.277]	0.301 [0.278, 0.325]	-0.032 [-0.045, -0.018]	0.212 [0.178, 0.247]
BabyLM-1.3B	Full context	0.287 [0.276, 0.298]	0.442 [0.418, 0.465]	-0.038 [-0.054, -0.022]	0.419 [0.385, 0.454]
	Attention-zeroed	0.205 [0.193, 0.217]	0.254 [0.230, 0.279]	-0.039 [-0.055, -0.024]	0.118 [0.083, 0.153]
GPT-2	Full context	0.236 [0.225, 0.247]	1.585 [1.542, 1.626]	0.006 [0.000, 0.011]	0.224 [0.204, 0.243]
	Attention-zeroed	0.258 [0.246, 0.269]	0.880 [0.838, 0.923]	-0.011 [-0.017, -0.005]	0.046 [0.026, 0.066]
GPT-2-M	Full context	0.231 [0.219, 0.242]	1.649 [1.606, 1.691]	0.004 [-0.001, 0.009]	0.206 [0.186, 0.226]
	Attention-zeroed	0.238 [0.227, 0.249]	0.949 [0.905, 0.992]	-0.007 [-0.013, -0.002]	0.053 [0.033, 0.073]
GPT-2-L	Full context	0.221 [0.210, 0.233]	1.718 [1.674, 1.760]	-0.001 [-0.006, 0.005]	0.197 [0.177, 0.216]
	Attention-zeroed	0.223 [0.212, 0.235]	1.010 [0.967, 1.053]	-0.004 [-0.009, 0.002]	0.056 [0.036, 0.075]
GPT-2-XL	Full context	0.219 [0.208, 0.230]	1.718 [1.676, 1.760]	0.000 [-0.005, 0.006]	0.175 [0.155, 0.195]
	Attention-zeroed	0.205 [0.194, 0.217]	1.028 [0.983, 1.071]	-0.007 [-0.013, -0.001]	0.065 [0.045, 0.086]
Pythia-160M	Full context	0.196 [0.185, 0.207]	1.676 [1.634, 1.718]	-0.010 [-0.016, -0.005]	0.237 [0.218, 0.256]
	Attention-zeroed	0.216 [0.205, 0.226]	0.964 [0.920, 1.010]	-0.008 [-0.013, -0.002]	0.038 [0.017, 0.058]
Pythia-410M	Full context	0.221 [0.211, 0.232]	1.736 [1.693, 1.777]	-0.003 [-0.008, 0.003]	0.171 [0.152, 0.191]
	Attention-zeroed	0.250 [0.239, 0.261]	0.940 [0.897, 0.983]	-0.008 [-0.013, -0.002]	0.052 [0.033, 0.072]
Pythia-1B	Full context	0.225 [0.214, 0.236]	1.730 [1.689, 1.773]	-0.002 [-0.008, 0.003]	0.155 [0.135, 0.175]
	Attention-zeroed	0.256 [0.245, 0.266]	0.923 [0.879, 0.967]	-0.013 [-0.019, -0.007]	0.064 [0.045, 0.084]
OLMo-1B	Full context	0.231 [0.220, 0.242]	1.708 [1.665, 1.749]	0.001 [-0.004, 0.007]	0.125 [0.105, 0.145]
	Attention-zeroed	0.233 [0.222, 0.244]	1.069 [1.025, 1.111]	-0.005 [-0.011, 0.001]	0.067 [0.047, 0.087]
OLMo-7B	Full context	0.245 [0.234, 0.256]	1.683 [1.641, 1.726]	-0.000 [-0.006, 0.005]	0.100 [0.080, 0.120]
	Attention-zeroed	0.242 [0.231, 0.254]	1.082 [1.039, 1.123]	-0.010 [-0.016, -0.004]	0.069 [0.049, 0.089]
OLMo-2-1B	Full context	0.224 [0.213, 0.235]	1.686 [1.644, 1.729]	-0.002 [-0.008, 0.003]	0.110 [0.090, 0.130]
	Attention-zeroed	0.237 [0.226, 0.248]	0.979 [0.935, 1.022]	-0.009 [-0.015, -0.003]	0.058 [0.038, 0.077]
OLMo-2-7B	Full context	0.249 [0.237, 0.260]	1.591 [1.546, 1.635]	-0.007 [-0.013, -0.001]	0.081 [0.060, 0.102]
	Attention-zeroed	0.243 [0.232, 0.254]	1.013 [0.970, 1.055]	-0.001 [-0.007, 0.005]	0.062 [0.042, 0.082]
Llama-1.3B	Full context	0.237 [0.226, 0.248]	1.660 [1.617, 1.703]	-0.001 [-0.006, 0.005]	0.114 [0.094, 0.134]
	Attention-zeroed	0.244 [0.234, 0.255]	1.055 [1.014, 1.097]	-0.009 [-0.015, -0.003]	0.060 [0.041, 0.080]
Llama-3-8B	Full context	0.252 [0.241, 0.263]	1.566 [1.522, 1.609]	-0.006 [-0.012, -0.001]	0.088 [0.068, 0.109]
	Attention-zeroed	0.226 [0.215, 0.237]	1.093 [1.050, 1.136]	-0.009 [-0.015, -0.003]	0.071 [0.050, 0.091]

Table 4: Frequency-dependent preference regression (Equation 1) for all 16 models at their final checkpoint, by attention condition. y_true and y_pred are z-scored within each fit; rel_freq is a proportion from -0.5 to $+0.5$ and log_freq is centered but not scaled, so both mean the same thing in every corpus. 95% credible intervals in brackets. Holistic storage predicts a negative y_pred × log_freq and a positive rel_freq × log_freq, both attenuating under attention-zeroing.

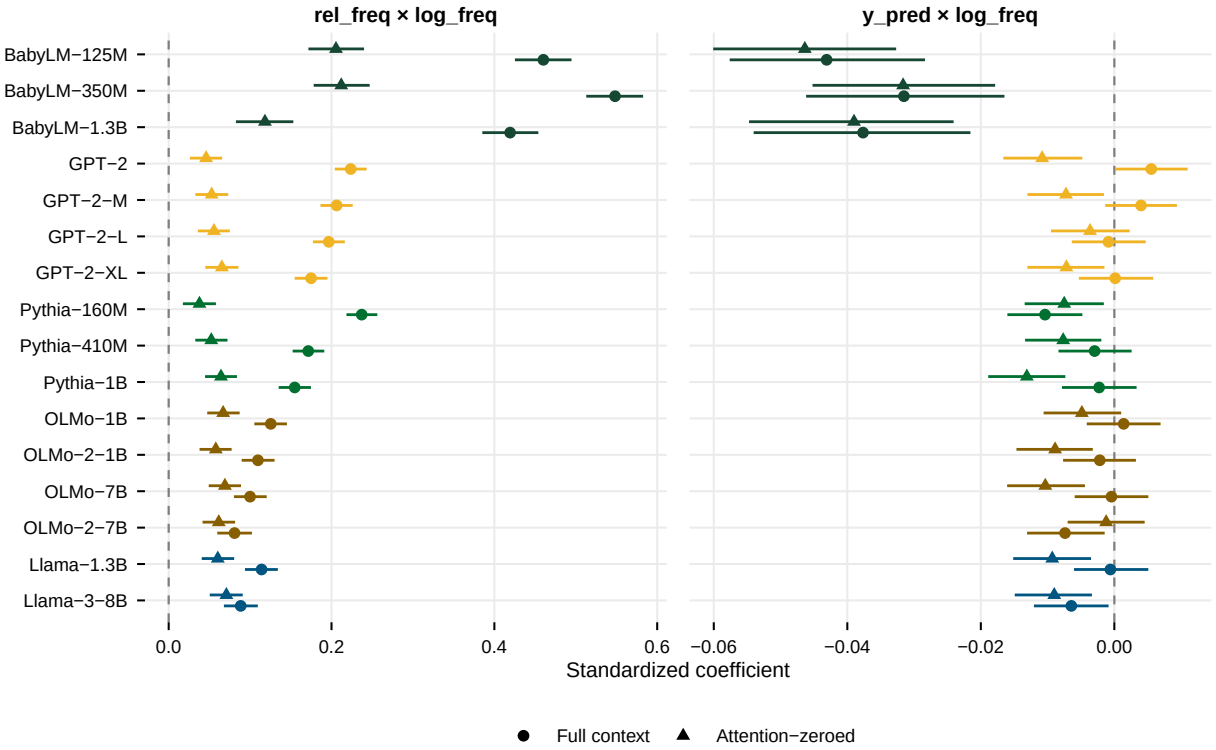


Figure 8: Moderation terms of Equation 1 by model and attention condition. Circles = full context; triangles = attention-zeroed. Points = posterior mean; bars = 95% credible intervals. Left: relative frequency matters more for frequent binomials, and attention-zeroing removes most of that. Right: abstract preferences matter less for frequent binomials, and attention-zeroing does not remove it.

Figure 8 shows the two moderation terms across models and conditions. The pattern is uniform across all sixteen: `rel_freq`×`log_freq` is credibly positive in every model under full context and much smaller under attention-zeroing, while `y_pred`×`log_freq` is credibly negative in every large-scale model under both conditions.

H Experiment 1: Supplementary Results

This appendix provides supplementary materials for Experiment 1 (§3), which examined final-layer probe performance for the three BabyLM models. Subsections report full R^2 tables with their label-shuffle controls, and a word-level replication. Coefficients for the frequency-dependent preference regression are reported together for all models in Appendix G.

H.1 Probe and Control R^2

Probe R^2 at the final hidden layer for the BabyLM models, at the binomial level by attention condition, alongside the control probe fit to randomly permuted labels on the same folds. ΔR^2 is the difference between the two.

Model	Condition	Split	R^2	Control R^2	ΔR^2
BabyLM-125M	Full context	B	0.129 (0.004)	0.0001 (0.0001)	0.129
		W	0.102 (0.027)	0.0003 (0.0006)	0.102
	Attention-zeroed	B	0.096 (0.005)	0.0000 (0.0000)	0.096
		W	0.080 (0.024)	0.0003 (0.0003)	0.080
BabyLM-350M	Full context	B	0.110 (0.005)	0.0000 (0.0001)	0.110
		W	0.093 (0.015)	0.0003 (0.0004)	0.093
	Attention-zeroed	B	0.070 (0.005)	0.0001 (0.0003)	0.070
		W	0.065 (0.015)	0.0006 (0.0007)	0.064

(continued)

Model	Condition	Split	R^2	Control R^2	ΔR^2
BabyLM-1.3B	Full context	B	0.068 (0.004)	0.0000 (0.0000)	0.068
		W	0.058 (0.011)	0.0002 (0.0003)	0.058
	Attention-zeroed	B	0.042 (0.005)	0.0000 (0.0000)	0.042
		W	0.038 (0.016)	0.0005 (0.0005)	0.038

Table 5: Probe R^2 at the final hidden layer for BabyLM models, at the binomial level by attention condition, with the label-shuffle control and ΔR^2 (R^2 minus control). B = binomial-held-out split; W = word-held-out split. SD across 10 folds in parentheses. Control values carry an extra decimal place because they sit near zero.

H.2 Cross-Model Agreement on Novel Binomials

A binomial absent from the training corpus cannot have been memorized (definitionally), but the model still assigns a preference, and that value is some deterministic function of the two words' representations. Such a function could reflect systematic knowledge, or it could be an arbitrary consequence of a particular set of weights. Probe generalization cannot distinguish these, because both are systematic functions of the input and both transfer to held-out words. Agreement between independently trained models can: an arbitrary function of one model's weights has no reason to match another model's preferences.

We therefore correlate y_{true} across models on identical binomials, separately for the attested and novel sets and for both attention conditions. As a baseline we shuffle one model's values, which removes any pairing between models while preserving both distributions.

The three BabyLM models are the decisive comparison. They share a training corpus and its ordering but differ in capacity, and they are the only models for which the novel binomials are genuinely absent from training (Section 4, Methods). On novel binomials they agree at $r = 0.50$ under full context and $r = 0.49$ under attention-zeroing, against a shuffled baseline of $r \approx 0.00$. In other words, preferences for binomials the models never encountered are reproducible across models of different sizes trained on the same data. They are therefore not arbitrary.

Because a Pearson correlation can in principle be carried by a minority of high-magnitude items, we checked whether the agreement is concentrated in a handful of binomials with strong preferences. This is not the case. Preferences on novel binomials are not weak to begin with: median $|y_{\text{true}}|$ ranges from 1.65 to 3.21 across the three models and both attention conditions.

Rank correlations track the Pearson values closely (0.36–0.55 for BabyLM on novel binomials), and the models order the same binomial the same way on 60% to 70% of items against a chance rate of 50%, a statistic that ignores magnitude and weights every binomial equally.

Two further patterns appear in Table 6. Agreement within a family is roughly constant across all four cells, between $r = 0.77$ and 0.81, and agreement between families is nearly as stable, between 0.62 and 0.67. Both hold up on novel binomials and under attention-zeroing: what the models share is confined neither to binomials they were trained on nor to models that share a training corpus. The within-family advantage nonetheless appears in every cell, which rules out the shared sentence frame as the source of the agreement: the frame is identical for all sixteen models, so it cannot produce a gap between within-family and between-family agreement.

Set	Condition	Scope	r	Shuffled
Attested	Attention-zeroed	All	0.640	-0.009
		Within family	0.767	-0.009
		Between family	0.616	-0.009
		Within BabyLM	0.520	-0.009
	Full context	All	0.691	0.004
		Within family	0.809	0.004
		Between family	0.669	0.004
		Within BabyLM	0.578	0.004

(continued)

Set	Condition	Scope	r	Shuffled
Novel	Attention-zeroed	All	0.647	-0.002
		Within family	0.778	-0.002
		Between family	0.623	-0.002
		Within BabyLM	0.486	-0.002
	Full context	All	0.658	-0.000
		Within family	0.787	-0.000
		Between family	0.634	-0.000
		Within BabyLM	0.496	-0.000

Table 6: Cross-model agreement (Pearson r on $y_{exttrue}$) for identical binomials, by item set and attention condition. Shuffled = one model's values permuted. Within-family averages over model pairs from the same family; BabyLM is broken out because its novel binomials are genuinely absent from those models' training corpus.

H.3 Word-Level Replication

We trained the MLP probe using word-level representations for the three BabyLM models to verify that the preference signal is not an artifact of mean pooling across the full binomial phrase. Under this condition the probe receives a concatenation of two word vectors extracted from isolated sentence contexts ("*the bread*", "*the butter*"), so it cannot rely on any co-occurrence signal from the two words' shared sentence context.

Binomial-held-out R^2 ranges from 0.07 to 0.13, matching the corresponding binomial-level values (0.07–0.13). The generalization pattern is preserved: word-held-out R^2 is consistently lower than binomial-held-out R^2 for all three models (word-held-out 0.06–0.09; binomial-minus-word gap 0.018–0.041), confirming that the lexical generalization cost seen at the binomial level is a property of the lexical representation itself rather than an artifact of the mean-pooled representation. The frequency-dependent preference regression was fit at the binomial level only, so it has no word-level counterpart. Full results are in Table 7.

Model	Mode	Split	R^2	Ctrl	ΔR^2
BabyLM-125M	Word-level	B	0.134 (0.005)	0.000 (0.000)	0.134
		W	0.093 (0.025)	0.000 (0.000)	0.093
BabyLM-350M		B	0.111 (0.004)	0.000 (0.000)	0.111
		W	0.086 (0.014)	0.000 (0.000)	0.086
BabyLM-1.3B		B	0.073 (0.004)	0.000 (0.000)	0.073
		W	0.055 (0.012)	0.000 (0.000)	0.055

Table 7: Word-level probe R^2 and ΔR^2 (main − control R^2) for BabyLM models (final layer). B = binomial-held-out split; W = word-held-out split. SD across 10 folds in parentheses.

I Experiment 2: Supplementary Results

This appendix provides supplementary materials for Experiment 2 (§4), which examined final-layer representations across all 16 models from five families. Subsections report full R^2 tables with their label-shuffle controls, and a word-level replication. Coefficients for the frequency-dependent preference regression are in Appendix G.

I.1 Probe and Control R^2

Probe R^2 for all 16 models at the final hidden layer, binomial-level input mode, by attention condition and cross-validation split, alongside the label-shuffle control fit on the same folds and the resulting ΔR^2. B = binomial-held-out; W = word-held-out. SD across 10 folds in parentheses.

Model	Condition	Split	R^2	Control R^2	ΔR^2
BabyLM-125M	Full context	B	0.129 (0.004)	0.0001 (0.0001)	0.129
		W	0.102 (0.027)	0.0003 (0.0006)	0.102
	Attention-zeroed	B	0.096 (0.005)	0.0000 (0.0000)	0.096
		W	0.080 (0.024)	0.0003 (0.0003)	0.080
BabyLM-350M	Full context	B	0.110 (0.005)	0.0000 (0.0001)	0.110
		W	0.093 (0.015)	0.0003 (0.0004)	0.093
	Attention-zeroed	B	0.070 (0.005)	0.0001 (0.0003)	0.070
		W	0.065 (0.015)	0.0006 (0.0007)	0.064
BabyLM-1.3B	Full context	B	0.068 (0.004)	0.0000 (0.0000)	0.068
		W	0.058 (0.011)	0.0002 (0.0003)	0.058
	Attention-zeroed	B	0.042 (0.005)	0.0000 (0.0000)	0.042
		W	0.038 (0.016)	0.0005 (0.0005)	0.038
GPT-2	Full context	B	0.181 (0.011)	0.0001 (0.0001)	0.181
		W	0.155 (0.046)	0.0005 (0.0006)	0.155
	Attention-zeroed	B	0.115 (0.006)	0.0000 (0.0000)	0.115
		W	0.106 (0.030)	0.0004 (0.0008)	0.105
GPT-2-M	Full context	B	0.184 (0.010)	0.0001 (0.0002)	0.184
		W	0.166 (0.044)	0.0008 (0.0010)	0.165
	Attention-zeroed	B	0.105 (0.006)	0.0001 (0.0001)	0.105
		W	0.094 (0.024)	0.0004 (0.0006)	0.093
GPT-2-L	Full context	B	0.184 (0.010)	0.0000 (0.0000)	0.184
		W	0.168 (0.052)	0.0005 (0.0007)	0.167
	Attention-zeroed	B	0.103 (0.009)	0.0001 (0.0001)	0.102
		W	0.091 (0.020)	0.0005 (0.0007)	0.091
GPT-2-XL	Full context	B	0.184 (0.010)	0.0000 (0.0001)	0.184
		W	0.169 (0.044)	0.0006 (0.0007)	0.169
	Attention-zeroed	B	0.100 (0.008)	0.0000 (0.0000)	0.100
		W	0.091 (0.027)	0.0004 (0.0007)	0.090
Pythia-160M	Full context	B	0.156 (0.010)	0.0001 (0.0001)	0.156
		W	0.140 (0.034)	0.0003 (0.0004)	0.140
	Attention-zeroed	B	0.087 (0.009)	0.0000 (0.0000)	0.087
		W	0.072 (0.022)	0.0004 (0.0006)	0.071
Pythia-410M	Full context	B	0.181 (0.008)	0.0000 (0.0000)	0.181
		W	0.161 (0.044)	0.0005 (0.0007)	0.161
	Attention-zeroed	B	0.097 (0.007)	0.0000 (0.0000)	0.097
		W	0.084 (0.017)	0.0008 (0.0012)	0.083
Pythia-1B	Full context	B	0.200 (0.008)	0.0000 (0.0000)	0.200
		W	0.186 (0.042)	0.0005 (0.0005)	0.186
	Attention-zeroed	B	0.107 (0.005)	0.0001 (0.0002)	0.107
		W	0.096 (0.020)	0.0003 (0.0003)	0.095
OLMo-1B	Full context	B	0.192 (0.008)	0.0000 (0.0001)	0.192
		W	0.181 (0.034)	0.0002 (0.0003)	0.180
	Attention-zeroed	B	0.108 (0.007)	0.0000 (0.0000)	0.108
		W	0.094 (0.028)	0.0001 (0.0001)	0.094
OLMo-7B	Full context	B	0.206 (0.009)	0.0001 (0.0001)	0.206
		W	0.191 (0.034)	0.0003 (0.0002)	0.190
	Attention-zeroed	B	0.115 (0.007)	0.0000 (0.0000)	0.115
		W	0.102 (0.032)	0.0003 (0.0005)	0.101
OLMo-2-1B	Full context	B	0.185 (0.010)	0.0000 (0.0000)	0.185
		W	0.174 (0.038)	0.0006 (0.0006)	0.173
	Attention-zeroed	B	0.096 (0.006)	0.0001 (0.0001)	0.096
		W	0.085 (0.022)	0.0005 (0.0009)	0.085

(continued)

Model	Condition	Split	R^2	Control R^2	ΔR^2
OLMo-2-7B	Full context	B	0.210 (0.008)	0.0000 (0.0000)	0.210
		W	0.197 (0.046)	0.0009 (0.0011)	0.197
	Attention-zeroed	B	0.114 (0.007)	0.0000 (0.0000)	0.114
		W	0.100 (0.025)	0.0004 (0.0006)	0.099
Llama-1.3B	Full context	B	0.193 (0.010)	0.0000 (0.0000)	0.193
		W	0.178 (0.042)	0.0004 (0.0008)	0.178
	Attention-zeroed	B	0.108 (0.008)	0.0001 (0.0001)	0.108
		W	0.092 (0.029)	0.0004 (0.0005)	0.092
Llama-3-8B	Full context	B	0.201 (0.006)	0.0001 (0.0001)	0.201
		W	0.190 (0.034)	0.0001 (0.0001)	0.190
	Attention-zeroed	B	0.110 (0.006)	0.0001 (0.0001)	0.110
		W	0.098 (0.033)	0.0006 (0.0005)	0.097

Table 8: Probe R^2 at the final hidden layer for all 16 models, at the binomial level by attention condition, with the label-shuffle control and ΔR^2 (R^2 minus control). B = binomial-held-out split; W = word-held-out split. SD across 10 folds in parentheses. Control values carry an extra decimal place because they sit near zero.

I.2 Word-Level Replication

Across all 16 models, word-level probing reproduces the main findings from Experiment 2. Binomial-held-out R^2 is slightly lower than the corresponding binomial-level values (by one to three percentage points), but rank order across families and the generalization pattern (binomial-held-out > word-held-out) are preserved in every case. For GPT-2, binomial-held-out R^2 under word-level probing ranges from 0.16 to 0.17 (vs. 0.18 at the binomial level); for Pythia, 0.15–0.18 (vs. 0.16–0.20); for OLMo, 0.17–0.18 (vs. 0.19–0.21); for Llama, 0.17–0.18 (vs. 0.19–0.20). The generalization gaps (binomial-held-out minus word-held-out) range from 0.019 to 0.030 across large-scale models, closely matching the corresponding binomial-level gaps (0.011–0.026). Full results are in Table 9.

Family	Model	Mode	Split	R^2	Ctrl	ΔR^2
BabyLM	BabyLM-125M	Word-level	B	0.134 (0.005)	0.000 (0.000)	0.134
			W	0.093 (0.025)	0.000 (0.000)	0.093
	BabyLM-350M		B	0.111 (0.004)	0.000 (0.000)	0.111
			W	0.086 (0.014)	0.000 (0.000)	0.086
	BabyLM-1.3B		B	0.073 (0.004)	0.000 (0.000)	0.073
			W	0.055 (0.012)	0.000 (0.000)	0.055
GPT-2	GPT-2		B	0.172 (0.006)	0.000 (0.000)	0.172
			W	0.142 (0.041)	0.001 (0.001)	0.141
	GPT-2-M		B	0.169 (0.007)	0.000 (0.000)	0.169
			W	0.146 (0.038)	0.001 (0.001)	0.145
	GPT-2-L		B	0.166 (0.006)	0.000 (0.000)	0.166
			W	0.145 (0.038)	0.000 (0.001)	0.145
	GPT-2-XL		B	0.164 (0.008)	0.000 (0.000)	0.164
			W	0.138 (0.041)	0.001 (0.001)	0.137
Llama	Llama-1.3B		B	0.173 (0.007)	0.000 (0.000)	0.173
			W	0.147 (0.047)	0.001 (0.001)	0.146
	Llama-3-8B		B	0.175 (0.007)	0.000 (0.000)	0.175
			W	0.153 (0.036)	0.000 (0.000)	0.153
OLMo	OLMo-1B		B	0.173 (0.008)	0.000 (0.000)	0.173
			W	0.152 (0.043)	0.000 (0.000)	0.151
	OLMo-7B		B	0.179 (0.008)	0.000 (0.000)	0.179
			W	0.159 (0.033)	0.000 (0.001)	0.159
	OLMo-2-1B		B	0.168 (0.006)	0.000 (0.000)	0.168
			W	0.141 (0.039)	0.000 (0.001)	0.141
	OLMo-2-7B		B	0.181 (0.007)	0.000 (0.000)	0.181
			W	0.163 (0.040)	0.000 (0.001)	0.162
Pythia	Pythia-160M		B	0.147 (0.007)	0.000 (0.000)	0.147
			W	0.127 (0.037)	0.000 (0.000)	0.127
	Pythia-410M		B	0.166 (0.006)	0.000 (0.000)	0.166
			W	0.138 (0.042)	0.000 (0.000)	0.138
	Pythia-1B		B	0.180 (0.006)	0.000 (0.000)	0.180
			W	0.153 (0.042)	0.000 (0.000)	0.153

(continued)

Family	Model	Mode	Split	R^2	Ctrl	ΔR^2

Table 9: Word-level probe R^2 and ΔR^2 (main − control R^2) for all 16 models (final layer). B = binomial-held-out split; W = word-held-out split. SD across 10 folds in parentheses.

J Experiment 3: Supplementary Results

This appendix provides supplementary materials for Experiment 3 (§5), which tracked probe R^2 and the frequency-dependent preference regression across training checkpoints for BabyLM and Pythia. Subsections report full numerical checkpoint tables and a word-level replication of the probe results.

J.1 Checkpoint Probe Tables

Full numerical results for the Experiment 3 checkpoint analysis. Tables report probe R^2 (binomial-held-out and word-held-out) and the coefficients of Equation 1 for BabyLM and Pythia across training checkpoints, for the binomial-level (mean-pooled) representation under full-context and attention-zeroed conditions.

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
BabyLM-125M	0	0 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		0 M	W	0.001 (0.001)	0.000 (0.000)	0.001
	24	20 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		20 M	W	0.001 (0.001)	0.000 (0.000)	0.001
	48	39 M	B	0.002 (0.001)	0.000 (0.000)	0.002
		39 M	W	0.002 (0.002)	0.000 (0.001)	0.002
	144	118 M	B	0.041 (0.005)	0.000 (0.000)	0.041
		118 M	W	0.028 (0.004)	0.001 (0.001)	0.028
	384	315 M	B	0.143 (0.009)	0.000 (0.000)	0.143
		315 M	W	0.129 (0.025)	0.000 (0.000)	0.129
	912	747 M	B	0.111 (0.007)	0.000 (0.000)	0.111
		747 M	W	0.082 (0.017)	0.001 (0.001)	0.082
	2280	1868 M	B	0.136 (0.006)	0.000 (0.000)	0.136
		1868 M	W	0.110 (0.028)	0.000 (0.001)	0.110
BabyLM-350M	0	0 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		0 M	W	0.002 (0.002)	0.000 (0.000)	0.001
	48	20 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		20 M	W	0.004 (0.003)	0.000 (0.001)	0.003
	96	39 M	B	0.001 (0.001)	0.000 (0.000)	0.001
		39 M	W	0.001 (0.002)	0.000 (0.000)	0.001
	288	118 M	B	0.098 (0.011)	0.000 (0.000)	0.098
		118 M	W	0.089 (0.011)	0.000 (0.000)	0.088
	768	315 M	B	0.139 (0.007)	0.000 (0.000)	0.139
		315 M	W	0.117 (0.021)	0.000 (0.000)	0.117
	1824	747 M	B	0.121 (0.004)	0.000 (0.000)	0.121
		747 M	W	0.098 (0.022)	0.001 (0.001)	0.097
	4560	1868 M	B	0.128 (0.006)	0.000 (0.000)	0.128
		1868 M	W	0.114 (0.018)	0.000 (0.000)	0.114
BabyLM-1.3B	0	0 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		0 M	W	0.000 (0.000)	0.001 (0.001)	-0.000
	97	20 M	B	0.001 (0.000)	0.000 (0.000)	0.000
		20 M	W	0.001 (0.001)	0.000 (0.001)	0.000
	194	40 M	B	0.001 (0.001)	0.000 (0.000)	0.001
		40 M	W	0.002 (0.002)	0.000 (0.000)	0.001
	582	119 M	B	0.095 (0.004)	0.000 (0.000)	0.095
		119 M	W	0.075 (0.018)	0.000 (0.001)	0.074
	1455	298 M	B	0.131 (0.008)	0.000 (0.000)	0.131
		298 M	W	0.101 (0.019)	0.000 (0.000)	0.101
	3686	755 M	B	0.133 (0.004)	0.000 (0.000)	0.133
		755 M	W	0.106 (0.026)	0.000 (0.000)	0.105
	9021	1848 M	B	0.102 (0.004)	0.000 (0.000)	0.102
		1848 M	W	0.088 (0.023)	0.000 (0.000)	0.088

Table 10: BabyLM checkpoint probe R^2:full context, binomial-level. B = binomial-held-out; W = word-held-out. SD in parentheses.

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
BabyLM-125M	0	0 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		0 M	W	0.000 (0.000)	0.000 (0.000)	-0.000
	24	20 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		20 M	W	0.000 (0.001)	0.000 (0.000)	0.000
	48	39 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		39 M	W	0.001 (0.001)	0.001 (0.001)	0.000
	144	118 M	B	0.008 (0.002)	0.000 (0.000)	0.008
		118 M	W	0.006 (0.003)	0.000 (0.000)	0.006
	384	315 M	B	0.074 (0.008)	0.000 (0.000)	0.074
		315 M	W	0.068 (0.011)	0.000 (0.000)	0.068
	912	747 M	B	0.090 (0.004)	0.000 (0.000)	0.090
		747 M	W	0.070 (0.012)	0.000 (0.000)	0.070
	2280	1868 M	B	0.106 (0.006)	0.000 (0.000)	0.105
		1868 M	W	0.092 (0.023)	0.001 (0.001)	0.091
BabyLM-350M	0	0 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		0 M	W	0.002 (0.003)	0.000 (0.000)	0.001
	48	20 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		20 M	W	0.000 (0.001)	0.001 (0.001)	-0.000
	96	39 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		39 M	W	0.001 (0.001)	0.001 (0.001)	0.000
	288	118 M	B	0.048 (0.008)	0.000 (0.000)	0.047
		118 M	W	0.038 (0.011)	0.000 (0.000)	0.037
	768	315 M	B	0.093 (0.008)	0.000 (0.000)	0.093
		315 M	W	0.081 (0.021)	0.000 (0.000)	0.080
	1824	747 M	B	0.110 (0.007)	0.000 (0.000)	0.110
		747 M	W	0.090 (0.023)	0.000 (0.000)	0.090
	4560	1868 M	B	0.095 (0.005)	0.000 (0.000)	0.095
		1868 M	W	0.082 (0.021)	0.001 (0.001)	0.081
BabyLM-1.3B	0	0 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		0 M	W	0.000 (0.000)	0.001 (0.001)	-0.001
	97	20 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		20 M	W	0.000 (0.000)	0.000 (0.001)	-0.000
	194	40 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		40 M	W	0.000 (0.001)	0.000 (0.001)	0.000
	582	119 M	B	0.062 (0.005)	0.000 (0.000)	0.062
		119 M	W	0.044 (0.016)	0.000 (0.000)	0.043
	1455	298 M	B	0.116 (0.009)	0.000 (0.000)	0.116
		298 M	W	0.101 (0.020)	0.000 (0.000)	0.101
	3686	755 M	B	0.114 (0.007)	0.000 (0.000)	0.114
		755 M	W	0.093 (0.026)	0.000 (0.001)	0.093
	9021	1848 M	B	0.072 (0.003)	0.000 (0.000)	0.071
		1848 M	W	0.066 (0.025)	0.001 (0.001)	0.065

Table 11: BabyLM checkpoint probe R^2:attention-zeroed, binomial-level.

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
Pythia-160M	0	0 M	B	0.013 (0.002)	0.000 (0.000)	0.013
		0 M	W	0.006 (0.004)	0.000 (0.001)	0.005
	16	34 M	B	0.006 (0.002)	0.000 (0.000)	0.006
		34 M	W	0.003 (0.003)	0.000 (0.000)	0.002
	32	67 M	B	0.004 (0.001)	0.000 (0.000)	0.004
		67 M	W	0.002 (0.003)	0.001 (0.001)	0.001
	64	134 M	B	0.037 (0.004)	0.000 (0.000)	0.037
		134 M	W	0.019 (0.006)	0.000 (0.000)	0.018
	256	537 M	B	0.112 (0.008)	0.000 (0.000)	0.112
		537 M	W	0.085 (0.020)	0.001 (0.001)	0.084
	512	1074 M	B	0.155 (0.006)	0.000 (0.000)	0.155
		1074 M	W	0.121 (0.014)	0.000 (0.000)	0.121
	1000	2097 M	B	0.153 (0.005)	0.000 (0.000)	0.153
		2097 M	W	0.120 (0.024)	0.000 (0.000)	0.120

(continued)

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
Pythia-410M	0	0 M	B	0.011 (0.002)	0.000 (0.000)	0.011
		0 M	W	0.005 (0.003)	0.001 (0.001)	0.005
	16	34 M	B	0.002 (0.001)	0.000 (0.000)	0.002
		34 M	W	0.001 (0.002)	0.000 (0.000)	0.001
	32	67 M	B	0.007 (0.001)	0.000 (0.000)	0.007
		67 M	W	0.005 (0.004)	0.001 (0.001)	0.004
	64	134 M	B	0.065 (0.005)	0.000 (0.000)	0.065
		134 M	W	0.052 (0.008)	0.000 (0.000)	0.051
	256	537 M	B	0.093 (0.007)	0.000 (0.000)	0.093
		537 M	W	0.071 (0.025)	0.000 (0.001)	0.071
	512	1074 M	B	0.159 (0.009)	0.000 (0.000)	0.159
		1074 M	W	0.132 (0.032)	0.000 (0.000)	0.132
	1000	2097 M	B	0.174 (0.006)	0.000 (0.000)	0.174
		2097 M	W	0.148 (0.027)	0.001 (0.002)	0.147
Pythia-1B	0	0 M	B	0.012 (0.003)	0.000 (0.000)	0.012
		0 M	W	0.003 (0.002)	0.001 (0.001)	0.002
	16	34 M	B	0.002 (0.001)	0.000 (0.000)	0.002
		34 M	W	0.002 (0.002)	0.000 (0.001)	0.002
	32	67 M	B	0.002 (0.001)	0.000 (0.000)	0.002
		67 M	W	0.001 (0.001)	0.000 (0.000)	0.000
	64	134 M	B	0.031 (0.004)	0.000 (0.000)	0.031
		134 M	W	0.015 (0.005)	0.001 (0.001)	0.015
	256	537 M	B	0.147 (0.008)	0.000 (0.000)	0.147
		537 M	W	0.106 (0.038)	0.001 (0.001)	0.106
	512	1074 M	B	0.169 (0.011)	0.000 (0.000)	0.168
		1074 M	W	0.121 (0.026)	0.001 (0.001)	0.120
	1000	2097 M	B	0.191 (0.006)	0.000 (0.000)	0.191
		2097 M	W	0.157 (0.023)	0.001 (0.001)	0.156

Table 12: Pythia checkpoint probe R^2:full context, binomial-level.

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
Pythia-160M	0	0 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		0 M	W	0.001 (0.001)	0.000 (0.001)	0.000
	16	34 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		34 M	W	0.001 (0.002)	0.000 (0.000)	0.001
	32	67 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		67 M	W	0.001 (0.002)	0.001 (0.001)	0.001
	64	134 M	B	0.009 (0.002)	0.000 (0.000)	0.009
		134 M	W	0.005 (0.004)	0.001 (0.001)	0.004
	256	537 M	B	0.071 (0.010)	0.000 (0.000)	0.071
		537 M	W	0.050 (0.018)	0.000 (0.000)	0.050
	512	1074 M	B	0.135 (0.007)	0.000 (0.000)	0.135
		1074 M	W	0.110 (0.023)	0.001 (0.001)	0.110
	1000	2097 M	B	0.157 (0.005)	0.000 (0.000)	0.157
		2097 M	W	0.130 (0.026)	0.001 (0.001)	0.129
Pythia-410M	0	0 M	B	0.001 (0.001)	0.000 (0.000)	0.001
		0 M	W	0.001 (0.001)	0.000 (0.000)	0.000
	16	34 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		34 M	W	0.001 (0.001)	0.000 (0.000)	0.000
	32	67 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		67 M	W	0.001 (0.002)	0.000 (0.001)	0.001
	64	134 M	B	0.037 (0.008)	0.000 (0.000)	0.037
		134 M	W	0.021 (0.009)	0.000 (0.000)	0.021
	256	537 M	B	0.063 (0.011)	0.000 (0.000)	0.063
		537 M	W	0.046 (0.012)	0.000 (0.001)	0.046
	512	1074 M	B	0.131 (0.007)	0.000 (0.000)	0.131
		1074 M	W	0.105 (0.028)	0.000 (0.000)	0.105
	1000	2097 M	B	0.178 (0.009)	0.000 (0.000)	0.177
		2097 M	W	0.154 (0.028)	0.000 (0.000)	0.153

(continued)

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
Pythia-1B	0	0 M	B	0.001 (0.001)	0.000 (0.000)	0.001
		0 M	W	0.001 (0.001)	0.000 (0.000)	0.001
	16	34 M	B	0.000 (0.001)	0.000 (0.000)	0.000
		34 M	W	0.002 (0.002)	0.000 (0.000)	0.001
	32	67 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		67 M	W	0.001 (0.002)	0.000 (0.000)	0.001
	64	134 M	B	0.017 (0.005)	0.000 (0.000)	0.017
		134 M	W	0.009 (0.006)	0.001 (0.003)	0.008
	256	537 M	B	0.102 (0.015)	0.000 (0.000)	0.102
		537 M	W	0.063 (0.026)	0.000 (0.001)	0.062
	512	1074 M	B	0.150 (0.009)	0.000 (0.000)	0.150
		1074 M	W	0.106 (0.020)	0.001 (0.000)	0.106
	1000	2097 M	B	0.163 (0.008)	0.000 (0.000)	0.163
		2097 M	W	0.136 (0.028)	0.000 (0.000)	0.136

Table 13: Pythia checkpoint probe R^2:attention-zeroed, binomial-level.

Model	Step	Condition	y_pred	rel_freq	y_pred × log_freq	rel_freq × log_freq
BabyLM-125M	0	Full context	0.019 [0.007, 0.030]	0.009 [-0.017, 0.035]	-0.006 [-0.024, 0.013]	0.062 [0.020, 0.104]
	24	Full context	0.024 [0.012, 0.036]	0.012 [-0.014, 0.037]	-0.003 [-0.021, 0.016]	0.060 [0.018, 0.104]
	48	Full context	0.052 [0.040, 0.064]	0.008 [-0.018, 0.033]	-0.010 [-0.028, 0.009]	0.056 [0.015, 0.098]
	144	Full context	0.198 [0.187, 0.210]	0.008 [-0.017, 0.033]	0.004 [-0.013, 0.021]	0.043 [0.003, 0.084]
	384	Full context	0.361 [0.350, 0.372]	0.165 [0.142, 0.188]	-0.020 [-0.035, -0.006]	0.103 [0.069, 0.137]
	912	Full context	0.331 [0.320, 0.342]	0.236 [0.212, 0.259]	-0.005 [-0.019, 0.009]	0.207 [0.172, 0.241]
	2280	Full context	0.374 [0.363, 0.385]	0.353 [0.330, 0.376]	-0.041 [-0.056, -0.027]	0.411 [0.376, 0.445]
	0	Attention-zeroed	0.009 [-0.002, 0.021]	0.004 [-0.021, 0.029]	0.004 [-0.014, 0.022]	0.062 [0.020, 0.104]
	24	Attention-zeroed	0.002 [-0.010, 0.013]	0.008 [-0.018, 0.033]	-0.000 [-0.018, 0.018]	0.061 [0.019, 0.103]
	48	Attention-zeroed	0.015 [0.003, 0.027]	-0.002 [-0.028, 0.024]	0.005 [-0.012, 0.023]	0.058 [0.016, 0.100]
	144	Attention-zeroed	0.105 [0.093, 0.117]	-0.009 [-0.034, 0.017]	0.017 [0.001, 0.034]	0.044 [0.002, 0.086]
	384	Attention-zeroed	0.270 [0.258, 0.281]	0.199 [0.175, 0.223]	-0.017 [-0.031, -0.003]	0.108 [0.072, 0.144]
	912	Attention-zeroed	0.292 [0.281, 0.303]	0.253 [0.229, 0.278]	-0.021 [-0.034, -0.007]	0.192 [0.159, 0.225]
	2280	Attention-zeroed	0.333 [0.322, 0.344]	0.258 [0.235, 0.281]	-0.041 [-0.054, -0.027]	0.219 [0.186, 0.252]
BabyLM-350M	0	Full context	0.029 [0.017, 0.040]	0.018 [-0.008, 0.043]	0.007 [-0.011, 0.025]	0.053 [0.011, 0.096]
	48	Full context	0.039 [0.027, 0.051]	0.016 [-0.010, 0.042]	0.002 [-0.017, 0.021]	0.063 [0.021, 0.106]
	96	Full context	0.038 [0.026, 0.050]	0.020 [-0.006, 0.046]	-0.001 [-0.019, 0.017]	0.060 [0.018, 0.102]
	288	Full context	0.293 [0.282, 0.304]	0.068 [0.044, 0.092]	-0.022 [-0.038, -0.006]	0.087 [0.049, 0.124]
	768	Full context	0.371 [0.360, 0.382]	0.178 [0.155, 0.202]	-0.025 [-0.040, -0.011]	0.135 [0.101, 0.170]
	1824	Full context	0.342 [0.331, 0.353]	0.309 [0.286, 0.333]	-0.034 [-0.049, -0.019]	0.331 [0.295, 0.366]
	4560	Full context	0.364 [0.354, 0.375]	0.447 [0.424, 0.470]	-0.035 [-0.049, -0.020]	0.510 [0.475, 0.545]
	0	Attention-zeroed	0.001 [-0.011, 0.013]	0.013 [-0.012, 0.038]	0.016 [-0.002, 0.034]	0.052 [0.010, 0.095]
	48	Attention-zeroed	0.014 [0.002, 0.026]	0.009 [-0.016, 0.035]	0.021 [0.003, 0.039]	0.064 [0.021, 0.107]
	96	Attention-zeroed	0.000 [-0.011, 0.012]	0.017 [-0.008, 0.042]	0.008 [-0.008, 0.026]	0.059 [0.017, 0.102]
	288	Attention-zeroed	0.218 [0.207, 0.230]	0.057 [0.032, 0.082]	-0.020 [-0.035, -0.005]	0.085 [0.047, 0.123]
	768	Attention-zeroed	0.326 [0.315, 0.337]	0.188 [0.164, 0.212]	-0.035 [-0.049, -0.020]	0.136 [0.102, 0.171]
	1824	Attention-zeroed	0.330 [0.319, 0.341]	0.256 [0.232, 0.280]	-0.044 [-0.059, -0.031]	0.216 [0.183, 0.250]
	4560	Attention-zeroed	0.307 [0.296, 0.318]	0.304 [0.280, 0.328]	-0.038 [-0.052, -0.024]	0.222 [0.189, 0.256]
BabyLM-1.3B	0	Full context	0.016 [0.004, 0.028]	0.016 [-0.010, 0.042]	0.013 [-0.005, 0.030]	0.066 [0.022, 0.110]
	97	Full context	0.042 [0.030, 0.053]	0.008 [-0.017, 0.033]	-0.000 [-0.018, 0.017]	0.065 [0.025, 0.106]
	194	Full context	0.045 [0.034, 0.057]	0.022 [-0.003, 0.048]	-0.011 [-0.029, 0.006]	0.059 [0.018, 0.099]
	582	Full context	0.284 [0.273, 0.295]	0.109 [0.086, 0.133]	-0.016 [-0.031, -0.001]	0.076 [0.040, 0.113]
	1455	Full context	0.353 [0.342, 0.364]	0.218 [0.195, 0.242]	-0.016 [-0.030, -0.001]	0.198 [0.164, 0.233]
	3686	Full context	0.354 [0.344, 0.365]	0.376 [0.353, 0.399]	-0.042 [-0.056, -0.028]	0.450 [0.416, 0.486]
	9021	Full context	0.324 [0.313, 0.335]	0.512 [0.489, 0.535]	-0.050 [-0.064, -0.035]	0.532 [0.498, 0.567]
	0	Attention-zeroed	-0.001 [-0.014, 0.010]	0.017 [-0.009, 0.042]	0.012 [-0.006, 0.030]	0.062 [0.018, 0.105]
	97	Attention-zeroed	0.018 [0.007, 0.030]	0.003 [-0.021, 0.028]	-0.010 [-0.028, 0.007]	0.063 [0.022, 0.104]
	194	Attention-zeroed	0.008 [-0.004, 0.020]	0.019 [-0.006, 0.045]	-0.005 [-0.022, 0.012]	0.057 [0.016, 0.099]
	582	Attention-zeroed	0.222 [0.211, 0.234]	0.109 [0.084, 0.133]	-0.017 [-0.032, -0.002]	0.075 [0.037, 0.111]
	1455	Attention-zeroed	0.329 [0.318, 0.340]	0.222 [0.199, 0.246]	-0.027 [-0.041, -0.013]	0.180 [0.147, 0.212]
	3686	Attention-zeroed	0.333 [0.322, 0.344]	0.269 [0.245, 0.292]	-0.039 [-0.052, -0.025]	0.215 [0.182, 0.247]
	9021	Attention-zeroed	0.256 [0.245, 0.267]	0.298 [0.275, 0.322]	-0.050 [-0.065, -0.036]	0.182 [0.149, 0.215]
Pythia-160M	0	Full context	0.068 [0.056, 0.080]	0.124 [0.077, 0.171]	-0.007 [-0.013, -0.000]	0.001 [-0.020, 0.022]
	16	Full context	0.051 [0.039, 0.063]	0.137 [0.092, 0.182]	-0.005 [-0.012, 0.001]	0.000 [-0.021, 0.021]
	32	Full context	0.046 [0.034, 0.058]	0.159 [0.114, 0.204]	-0.002 [-0.009, 0.004]	-0.000 [-0.022, 0.021]
	64	Full context	0.169 [0.157, 0.180]	0.194 [0.149, 0.239]	0.010 [0.003, 0.016]	0.002 [-0.019, 0.023]
	256	Full context	0.276 [0.265, 0.287]	0.560 [0.518, 0.602]	0.001 [-0.005, 0.007]	0.015 [-0.004, 0.035]
	512	Full context	0.341 [0.330, 0.352]	0.787 [0.745, 0.829]	-0.003 [-0.009, 0.004]	0.077 [0.058, 0.096]
	1000	Full context	0.310 [0.299, 0.321]	0.995 [0.953, 1.037]	-0.015 [-0.021, -0.009]	0.173 [0.154, 0.193]
	0	Attention-zeroed	0.020 [0.008, 0.032]	0.123 [0.078, 0.168]	0.001 [-0.006, 0.007]	0.001 [-0.020, 0.023]
	16	Attention-zeroed	0.015 [0.003, 0.026]	0.137 [0.092, 0.182]	-0.001 [-0.007, 0.006]	0.001 [-0.021, 0.022]
	32	Attention-zeroed	0.014 [0.003, 0.026]	0.164 [0.117, 0.210]	0.001 [-0.005, 0.007]	-0.001 [-0.022, 0.020]
	64	Attention-zeroed	0.108 [0.096, 0.120]	0.184 [0.138, 0.230]	0.000 [-0.006, 0.007]	0.002 [-0.019, 0.023]
	256	Attention-zeroed	0.231 [0.219, 0.242]	0.551 [0.507, 0.595]	-0.001 [-0.007, 0.006]	0.019 [-0.001, 0.040]
	512	Attention-zeroed	0.286 [0.274, 0.297]	0.859 [0.816, 0.903]	-0.010 [-0.016, -0.004]	0.052 [0.033, 0.071]
	1000	Attention-zeroed	0.332 [0.321, 0.343]	0.789 [0.748, 0.831]	-0.011 [-0.017, -0.005]	0.052 [0.033, 0.072]

(continued)

Model	Step	Condition	y_pred	rel_freq	y_pred × log_freq	rel_freq × log_freq
Pythia-410M	0	Full context	0.088 [0.076, 0.100]	0.114 [0.068, 0.159]	0.001 [-0.005, 0.007]	0.004 [-0.017, 0.025]
	16	Full context	0.046 [0.034, 0.058]	0.143 [0.098, 0.188]	-0.005 [-0.011, 0.001]	0.004 [-0.017, 0.025]
	32	Full context	0.116 [0.105, 0.128]	0.157 [0.111, 0.201]	-0.006 [-0.013, 0.000]	0.004 [-0.017, 0.025]
	64	Full context	0.252 [0.241, 0.264]	0.216 [0.171, 0.260]	0.008 [0.001, 0.014]	0.007 [-0.013, 0.028]
	256	Full context	0.251 [0.240, 0.262]	0.507 [0.463, 0.551]	0.002 [-0.003, 0.008]	0.025 [0.004, 0.045]
	512	Full context	0.343 [0.332, 0.353]	0.705 [0.663, 0.747]	-0.002 [-0.008, 0.004]	0.047 [0.028, 0.066]
	1000	Full context	0.326 [0.315, 0.337]	0.914 [0.872, 0.957]	-0.012 [-0.018, -0.006]	0.156 [0.137, 0.176]
	0	Attention-zeroed	0.011 [-0.001, 0.023]	0.119 [0.073, 0.165]	0.002 [-0.005, 0.008]	0.003 [-0.018, 0.024]
	16	Attention-zeroed	0.013 [0.002, 0.025]	0.144 [0.099, 0.189]	-0.003 [-0.010, 0.003]	0.003 [-0.018, 0.024]
	32	Attention-zeroed	0.034 [0.022, 0.046]	0.161 [0.115, 0.207]	0.001 [-0.006, 0.007]	0.003 [-0.017, 0.024]
	64	Attention-zeroed	0.204 [0.193, 0.216]	0.189 [0.144, 0.234]	0.002 [-0.005, 0.008]	0.015 [-0.006, 0.036]
	256	Attention-zeroed	0.207 [0.196, 0.219]	0.492 [0.449, 0.535]	-0.004 [-0.011, 0.002]	0.025 [0.005, 0.045]
	512	Attention-zeroed	0.293 [0.282, 0.304]	0.776 [0.735, 0.817]	-0.004 [-0.010, 0.002]	0.036 [0.017, 0.056]
	1000	Attention-zeroed	0.337 [0.326, 0.348]	0.797 [0.755, 0.839]	-0.010 [-0.016, -0.005]	0.052 [0.032, 0.071]
Pythia-1B	0	Full context	0.106 [0.095, 0.118]	0.146 [0.101, 0.191]	0.001 [-0.005, 0.007]	-0.007 [-0.028, 0.014]
	16	Full context	0.011 [-0.001, 0.023]	0.161 [0.114, 0.207]	-0.003 [-0.009, 0.003]	-0.001 [-0.022, 0.020]
	32	Full context	0.031 [0.019, 0.042]	0.181 [0.135, 0.226]	-0.005 [-0.011, 0.002]	0.003 [-0.018, 0.024]
	64	Full context	0.153 [0.141, 0.164]	0.294 [0.249, 0.338]	-0.003 [-0.009, 0.003]	0.020 [-0.000, 0.040]
	256	Full context	0.328 [0.318, 0.339]	0.610 [0.568, 0.652]	0.005 [-0.001, 0.011]	0.033 [0.013, 0.053]
	512	Full context	0.355 [0.345, 0.366]	0.751 [0.708, 0.794]	-0.007 [-0.013, -0.001]	0.092 [0.072, 0.111]
	1000	Full context	0.327 [0.315, 0.338]	1.031 [0.989, 1.073]	-0.009 [-0.015, -0.003]	0.180 [0.161, 0.199]
	0	Attention-zeroed	0.009 [-0.004, 0.020]	0.153 [0.107, 0.198]	-0.007 [-0.013, -0.000]	-0.008 [-0.029, 0.012]
	16	Attention-zeroed	0.007 [-0.005, 0.019]	0.162 [0.116, 0.207]	-0.007 [-0.013, -0.001]	-0.001 [-0.023, 0.020]
	32	Attention-zeroed	0.015 [0.003, 0.027]	0.181 [0.135, 0.228]	-0.001 [-0.008, 0.005]	0.003 [-0.018, 0.024]
	64	Attention-zeroed	0.109 [0.097, 0.121]	0.283 [0.238, 0.329]	-0.006 [-0.012, 0.001]	0.024 [0.003, 0.045]
	256	Attention-zeroed	0.271 [0.260, 0.282]	0.648 [0.605, 0.692]	0.001 [-0.005, 0.007]	0.036 [0.017, 0.056]
	512	Attention-zeroed	0.287 [0.276, 0.298]	0.849 [0.808, 0.892]	-0.009 [-0.015, -0.003]	0.062 [0.043, 0.082]
	1000	Attention-zeroed	0.318 [0.307, 0.329]	0.863 [0.820, 0.905]	-0.010 [-0.016, -0.004]	0.050 [0.031, 0.070]

Table 14: Frequency-dependent preference regression (Equation 1) at every training checkpoint, both attention conditions. `y_true` and `y_pred` are z-scored within each fit; `rel_freq` is a proportion from -0.5 to $+0.5$ and `log_freq` is centered but not scaled, so both mean the same thing in every corpus. 95% credible intervals in brackets.

J.2 Word-Level Replication

Word-level probe results across BabyLM and Pythia checkpoints replicate the key patterns from the binomial-level analysis. The generalization gap (binomial-held-out minus word-held-out) is absent at the earliest checkpoints, where neither split is distinguishable from zero, and widens over the course of training. For BabyLM, the gap is within rounding of zero at the first checkpoint and reaches 0.026–0.042 by the final one, the same qualitative trajectory observed at the binomial level. Full R^2 results are in Table 15 and Table 16. The frequency-dependent preference regression was fit at the binomial level only, so it has no word-level counterpart here; its checkpoint coefficients are in Table 14.

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
BabyLM-125M	0	0 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		0 M	W	0.001 (0.001)	0.000 (0.000)	0.001
	24	20 M	B	0.001 (0.000)	0.000 (0.000)	0.000
		20 M	W	0.001 (0.002)	0.000 (0.000)	0.001
	48	39 M	B	0.002 (0.001)	0.000 (0.000)	0.002
		39 M	W	0.003 (0.003)	0.000 (0.000)	0.003
	144	118 M	B	0.088 (0.008)	0.000 (0.000)	0.088
		118 M	W	0.077 (0.017)	0.001 (0.001)	0.076
	384	315 M	B	0.206 (0.009)	0.000 (0.000)	0.206
		315 M	W	0.184 (0.019)	0.001 (0.001)	0.183
	912	747 M	B	0.140 (0.005)	0.000 (0.000)	0.140
		747 M	W	0.101 (0.028)	0.000 (0.000)	0.100
	2280	1868 M	B	0.143 (0.005)	0.000 (0.000)	0.143
		1868 M	W	0.105 (0.027)	0.001 (0.001)	0.104

(continued)

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
BabyLM-350M	0	0 M	B	0.001 (0.000)	0.000 (0.000)	0.000
		0 M	W	0.000 (0.000)	0.001 (0.001)	-0.000
	48	20 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		20 M	W	0.001 (0.001)	0.000 (0.001)	0.001
	96	39 M	B	0.001 (0.000)	0.000 (0.000)	0.001
		39 M	W	0.002 (0.001)	0.000 (0.001)	0.002
	288	118 M	B	0.154 (0.009)	0.000 (0.000)	0.154
		118 M	W	0.143 (0.018)	0.000 (0.001)	0.143
	768	315 M	B	0.181 (0.007)	0.000 (0.000)	0.181
		315 M	W	0.149 (0.025)	0.001 (0.001)	0.148
	1824	747 M	B	0.131 (0.006)	0.000 (0.000)	0.131
		747 M	W	0.091 (0.020)	0.001 (0.001)	0.090
	4560	1868 M	B	0.131 (0.006)	0.000 (0.000)	0.131
		1868 M	W	0.105 (0.015)	0.001 (0.001)	0.104
BabyLM-1.3B	0	0 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		0 M	W	0.001 (0.001)	0.000 (0.000)	0.000
	97	20 M	B	0.000 (0.000)	0.000 (0.000)	0.000
		20 M	W	0.000 (0.000)	0.001 (0.000)	-0.000
	194	40 M	B	0.002 (0.001)	0.000 (0.000)	0.001
		40 M	W	0.002 (0.002)	0.000 (0.001)	0.001
	582	119 M	B	0.138 (0.007)	0.000 (0.000)	0.138
		119 M	W	0.117 (0.024)	0.000 (0.001)	0.117
	1455	298 M	B	0.155 (0.006)	0.000 (0.000)	0.155
		298 M	W	0.123 (0.021)	0.001 (0.001)	0.122
	3686	755 M	B	0.138 (0.006)	0.000 (0.000)	0.138
		755 M	W	0.097 (0.022)	0.001 (0.001)	0.096
	9021	1848 M	B	0.106 (0.005)	0.000 (0.000)	0.106
		1848 M	W	0.080 (0.012)	0.000 (0.000)	0.080

Table 15: BabyLM checkpoint probe R^2:full context, word-level. B = binomial-held-out; W = word-held-out. SD in parentheses.

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
Pythia-160M	0	0 M	B	0.055 (0.004)	0.000 (0.000)	0.055
		0 M	W	0.024 (0.009)	0.000 (0.001)	0.023
	16	34 M	B	0.027 (0.003)	0.000 (0.000)	0.027
		34 M	W	0.009 (0.004)	0.001 (0.001)	0.008
	32	67 M	B	0.013 (0.003)	0.000 (0.000)	0.013
		67 M	W	0.004 (0.002)	0.001 (0.001)	0.003
	64	134 M	B	0.088 (0.008)	0.000 (0.000)	0.088
		134 M	W	0.056 (0.018)	0.000 (0.001)	0.056
	256	537 M	B	0.172 (0.010)	0.000 (0.000)	0.172
		537 M	W	0.123 (0.032)	0.000 (0.000)	0.123
	512	1074 M	B	0.188 (0.006)	0.000 (0.000)	0.188
		1074 M	W	0.133 (0.021)	0.000 (0.000)	0.132
	1000	2097 M	B	0.159 (0.004)	0.000 (0.000)	0.159
		2097 M	W	0.117 (0.024)	0.000 (0.001)	0.117
Pythia-410M	0	0 M	B	0.054 (0.005)	0.000 (0.000)	0.053
		0 M	W	0.024 (0.009)	0.000 (0.000)	0.024
	16	34 M	B	0.014 (0.002)	0.000 (0.000)	0.014
		34 M	W	0.004 (0.003)	0.001 (0.001)	0.003
	32	67 M	B	0.034 (0.004)	0.000 (0.000)	0.034
		67 M	W	0.018 (0.011)	0.000 (0.001)	0.018
	64	134 M	B	0.127 (0.007)	0.000 (0.000)	0.127
		134 M	W	0.101 (0.016)	0.000 (0.000)	0.101
	256	537 M	B	0.134 (0.009)	0.000 (0.000)	0.134
		537 M	W	0.102 (0.022)	0.001 (0.001)	0.101
	512	1074 M	B	0.199 (0.007)	0.000 (0.000)	0.199
		1074 M	W	0.151 (0.033)	0.000 (0.001)	0.151
	1000	2097 M	B	0.179 (0.007)	0.000 (0.000)	0.179
		2097 M	W	0.137 (0.031)	0.000 (0.000)	0.137

(continued)

Model	Step	Tokens	Split	R^2	Ctrl	ΔR^2
Pythia-1B	0	0 M	B	0.061 (0.005)	0.000 (0.000)	0.061
		0 M	W	0.018 (0.005)	0.000 (0.000)	0.018
	16	34 M	B	0.016 (0.003)	0.000 (0.000)	0.016
		34 M	W	0.005 (0.003)	0.000 (0.001)	0.004
	32	67 M	B	0.013 (0.002)	0.000 (0.000)	0.013
		67 M	W	0.004 (0.002)	0.000 (0.000)	0.003
	64	134 M	B	0.075 (0.005)	0.000 (0.000)	0.075
		134 M	W	0.042 (0.015)	0.001 (0.001)	0.042
	256	537 M	B	0.205 (0.010)	0.000 (0.000)	0.205
		537 M	W	0.137 (0.037)	0.001 (0.001)	0.137
	512	1074 M	B	0.211 (0.005)	0.000 (0.000)	0.211
		1074 M	W	0.147 (0.030)	0.000 (0.000)	0.147
	1000	2097 M	B	0.197 (0.005)	0.000 (0.000)	0.197
		2097 M	W	0.143 (0.024)	0.001 (0.000)	0.142

Table 16: Pythia checkpoint probe R^2:full context, word-level. B = binomial-held-out; W = word-held-out. SD in parentheses.

K Tokenization Artifact at Initialization

At step 0 the untrained Pythia models show a small but credible effect of relative frequency (Table 17). This appendix shows that the effect is an artifact of subword tokenization rather than any ordering preference in the untrained networks, and that removing it leaves the trained models unchanged.

Where a binomial's left context ends in a quotation mark, parenthesis, or hyphen rather than whitespace, whichever word comes first loses its leading space, and byte-pair encoding segments many words differently without one. The two orderings then sum different numbers of span tokens. At initialization every token costs roughly the log of the vocabulary size, so a one-token difference is worth about 11 nats for Pythia, swamping a typical ordering preference of well under one. This affects 571 of the 27,835 corpus binomials (2.1%): 53% of those whose context ends in a quotation mark against 0.1% of those ending in whitespace.

We refit Equation 1 for the three Pythia models at step 0 and at their final checkpoint, excluding those 571 items. Everything else is held fixed, and the scaling is recomputed on the retained rows. Table 17 gives the relative-frequency coefficient before and after. At step 0 the effect is credible in all six cells as reported and in none of them once the items are dropped. At the final checkpoint the coefficient moves by at most 0.007, which is well inside its credible interval. The artifact dominates at step 0 only because there is no real signal competing with it: those 571 items carry roughly 80% of the variance in an untrained model's preferences and about 3% in a trained one.

Checkpoint	Model	Condition	As reported	Excluded
Step 0	Pythia-160M	Attention-zeroed	+0.123 [+0.078, +0.168]	-0.032 [-0.078, +0.012]
		Full context	+0.124 [+0.077, +0.171]	-0.027 [-0.072, +0.017]
	Pythia-410M	Attention-zeroed	+0.119 [+0.073, +0.165]	-0.047 [-0.093, -0.002]
		Full context	+0.114 [+0.068, +0.159]	-0.052 [-0.097, -0.008]
	Pythia-1B	Attention-zeroed	+0.153 [+0.107, +0.198]	+0.022 [-0.024, +0.067]
		Full context	+0.146 [+0.101, +0.191]	+0.009 [-0.037, +0.054]
Final	Pythia-160M	Attention-zeroed	+0.964 [+0.920, +1.010]	+0.957 [+0.914, +1.000]
		Full context	+1.676 [+1.634, +1.718]	+1.683 [+1.641, +1.725]
	Pythia-410M	Attention-zeroed	+0.940 [+0.897, +0.983]	+0.936 [+0.894, +0.980]
		Full context	+1.736 [+1.693, +1.777]	+1.734 [+1.691, +1.776]
	Pythia-1B	Attention-zeroed	+0.923 [+0.879, +0.967]	+0.919 [+0.876, +0.962]
		Full context	+1.730 [+1.689, +1.773]	+1.725 [+1.681, +1.768]

Table 17: Relative-frequency coefficient of Equation 1 for the Pythia models, as reported and with the 571 asymmetrically tokenized binomials excluded. Posterior mean with 95% credible interval. The step-0 effect disappears; the final-checkpoint effect is unchanged.

L Source Timing by Model

Per-model values behind the timing comparison in Experiment 3, for the full-context condition. Both criteria are defined over the checkpoint fits of Equation 1, so the two sources are measured on the same scale: *detectable* is the first checkpoint whose posterior sits credibly above that coefficient's own step-0 value and stays there, and *three quarters* is the first checkpoint at which the coefficient has covered three quarters of its movement from step 0 to its final value. Both are reported as cumulative training tokens. The checkpoint grid is coarse and log-spaced, so equal entries mean the ordering is unresolved at this spacing rather than simultaneous. Appendix K covers why step 0 rather than zero is the baseline, and the Limitations section covers why the two sources are not on equal footing.

Source	Model	Detectable	Three quarters
Abstract preferences	BabyLM-125M	39	315
	BabyLM-350M	118	118
	BabyLM-1.3B	20	119
	Pythia-160M	134	537
	Pythia-410M	67	1074
	Pythia-1B	134	537
Relative frequency	BabyLM-125M	315	1868
	BabyLM-350M	118	1868
	BabyLM-1.3B	119	1848
	Pythia-160M	134	1074
	Pythia-410M	134	2097
	Pythia-1B	134	2097
Rel. freq. $imes$ frequency	BabyLM-125M	747	1868
	BabyLM-350M	315	1868
	BabyLM-1.3B	298	755
	Pythia-160M	1074	2097
	Pythia-410M	1074	2097
	Pythia-1B	537	2097

Table 18: When each source of Equation 1 becomes measurable, in millions of cumulative training tokens, full context. Detectable = credibly above the same coefficient's step-0 value and staying there. Three quarters = first checkpoint covering three quarters of that coefficient's change from step 0 to final.